

01 GET /alarms

RULES ["compatibility aliases derive only from canonical fields", "source R03 requires alarm_type OOS, rule_code R03_CONSEC, and value null", "source SUMMARY requires rule_code SUMMARY_OOC", "source TRACE requires rule_code TRACE_OOS"]

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {
      "area": {
        "nullable": true,
        "required": false,
        "schema": {
          "enum": ["ALL", "Etch", "Photo"],
          "type": "string"
        },
        "chamber": {
          "nullable": true,
          "required": false,
          "schema": {
            "type": "string"
          },
          "date_from": {
            "nullable": true,
            "required": false,
            "schema": {
              "format": "date",
              "type": "string"
            },
            "date_to": {
            "nullable": true,
            "required": false,
            "schema": {
              "format": "date",
              "type": "string"
            },
            "equipment": {
              "nullable": true,
              "required": false,
              "schema": {
                "type": "string"
              },
              "include_derived": {
                "nullable": false,
                "required": false,
                "schema": {
                  "default": false,
                  "type": "boolean"
                },
                "parameter": {
                  "nullable": true,
                  "required": false,
                  "schema": {
                    "type": "string"
                  },
                  "source": {
                    "nullable": true,
                    "required": true,
                    "schema": {
                      "enum": ["R03", "SUMMARY", "TRACE"],
                      "type": "string"
                    },
                    "responses": {
                      "200": {
                        "schema": {
                          "items": {
                            "additional_properties": false,
                            "fields": {
                              "action_code": {
                                "nullable": true,
                                "required": false,
                                "schema": {
                                  "enum": ["EQP_HOLD", "MONITORING", "WARNING"],
                                  "type": "string"
                                },
                              "alarm_id": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "min_length": 1,
                                  "type": "string"
                                },
                              "alarm_type": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "enum": ["OOC", "OOS"],
                                  "type": "string"
                                },
                              "area": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "enum": ["Etch", "Photo"],
                                  "type": "string"
                                },
                              "chamber": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "min_length": 1,
                                  "type": "string"
                                },
                              "chamber_id": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "min_length": 1,
                                  "type": "string"
                                },
                              "cl": {
                                "nullable": true,
                                "required": false,
                                "schema": {
                                  "type": "number"
                                },
                              "equipment": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "min_length": 1,
                                  "type": "string"
                                },
                              "equipment_id": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "min_length": 1,
                                  "type": "string"
                                },
                              "fault": {
                                "nullable": true,
                                "required": false,
                                "schema": {
                                  "enum": ["FOC", "MFD", "OTH", "RFM", "TMD"],
                                  "type": "string"
                                },
                              "lcl": {
                                "nullable": true,
                                "required": false,
                                "schema": {
                                  "type": "string"
                                },
                              "lot": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "min_length": 1,
                                  "type": "string"
                                },
                              "lot_id": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "min_length": 1,
                                  "type": "string"
                                },
                              "mes": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "type": "string"
                                },
                              "mes_status": {
                                "nullable": true,
                                "required": false,
                                "schema": {
                                  "enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"],
                                  "type": "string"
                                },
                              "notify": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "type": "boolean"
                                },
                              "notify_status": {
                                "nullable": true,
                                "required": false,
                                "schema": {
                                  "enum": ["FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"],
                                  "type": "string"
                                },
                              "occurred_at": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "format": "date-time",
                                  "type": "string"
                                },
                              "parameter": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "min_length": 1,
                                  "type": "string"
                                },
                              "parameter_id": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "min_length": 1,
                                  "type": "string"
                                },
                              "predicted_fault_code": {
                                "nullable": true,
                                "required": false,
                                "schema": {
                                  "enum": ["FOC", "MFD", "OTH", "RFM", "TMD"],
                                  "type": "string"
                                },
                              "recipe": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "min_length": 1,
                                  "type": "string"
                                },
                              "recipe_id": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "min_length": 1,
                                  "type": "string"
                                },
                              "recipe_step_no": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "minimum": 1.0,
                                  "type": "integer"
                                },
                              "rule_code": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "enum": ["R03_CONSEC", "SUMMARY_OOC", "TRACE_OOS"],
                                  "type": "string"
                                },
                              "seq_no": {
                                "nullable": true,
                                "required": false,
                                "schema": {
                                  "minimum": 0.0,
                                  "type": "integer"
                                },
                              "source": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "enum": ["R03", "SUMMARY", "TRACE"],
                                  "type": "string"
                                },
                              "statistic_type": {
                                "nullable": true,
                                "required": false,
                                "schema": {
                                  "type": "string"
                                },
                              "step_no": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "minimum": 1.0,
                                  "type": "integer"
                                },
                              "ucl": {
                                "nullable": true,
                                "required": false,
                                "schema": {
                                  "type": "number"
                                },
                              "value": {
                                "nullable": true,
                                "required": false,
                                "schema": {
                                  "type": "number"
                                },
                              "wafer": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "min_length": 1,
                                  "type": "string"
                                },
                              "wafer_id": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "min_length": 1,
                                  "type": "string"
                                },
                              "type": "object",
                              "type": "array",
                              "shape": "array",
                              "422": {
                                "schema": {
                                  "additional_properties": true,
                                  "fields": {
                                    "detail": {
                                      "nullable": false,
                                      "required": false,
                                      "schema": {
                                        "items": {
                                          "additional_properties": true,
                                          "fields": {
                                            "loc": {
                                              "nullable": false,
                                              "required": true,
                                              "schema": {
                                                "items": {
                                                  "type": "union",
                                                  "variants": [
                                                    {
                                                      "type": "string"
                                                    },
                                                    {
                                                      "type": "integer"
                                                    }
                                                  ],
                                                  "type": "array"
                                                },
                                                "msg": {
                                                  "nullable": false,
                                                  "required": true,
                                                  "schema": {
                                                    "type": "string"
                                                  },
                                                  "type": {
                                                    "nullable": false,
                                                    "required": true,
                                                    "schema": {
                                                      "type": "string"
                                                    },
                                                  "type": {
                                                    "nullable": false,
                                                    "required": true,
                                                    "schema": {
                                                      "type": "string"
                                                    },
                                                  "type": "object",
                                                  "type": "array",
                                                  "shape": "object"
                                                }
                                              }
                                            }
                                          }
                                        }
                                      }
                                    }
                                  }
                                }
                              }
                            }
                          }
                        }
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

02 GET /trace

RULES []

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {
      "chamber": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
      "lot": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
      "parameter": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
      "wafer": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      }
    },
    "responses": {
      "200": {
        "schema": {
          "items": {
            "additional_properties": false,
            "fields": {
              "measured_at": {
                "nullable": false,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
              "recipe_step_no": {
                "nullable": false,
                "required": true,
                "schema": {
                  "minimum": 1.0,
                  "type": "integer"
                },
              "seq_no": {
                "nullable": false,
                "required": true,
                "schema": {
                  "minimum": 0.0,
                  "type": "integer"
                },
              "value": {
                "nullable": false,
                "required": true,
                "schema": {
                  "type": "number"
                }
              },
              "type": "object",
              "type": "array",
              "shape": "array",
              "422": {
                "schema": {
                  "additional_properties": true,
                  "fields": {
                    "detail": {
                      "nullable": false,
                      "required": false,
                      "schema": {
                        "items": {
                          "additional_properties": true,
                          "fields": {
                            "loc": {
                              "nullable": false,
                              "required": true,
                              "schema": {
                                "items": {
                                  "type": "union",
                                  "variants": [
                                    {
                                      "type": "string"
                                    },
                                    {
                                      "type": "integer"
                                    }
                                  ],
                                  "type": "array"
                                },
                                "msg": {
                                  "nullable": false,
                                  "required": true,
                                  "schema": {
                                    "type": "string"
                                  },
                                  "type": {
                                    "nullable": false,
                                    "required": true,
                                    "schema": {
                                      "type": "string"
                                    },
                                  "type": {
                                    "nullable": false,
                                    "required": true,
                                    "schema": {
                                      "type": "string"
                                    },
                                  "type": "object",
                                  "type": "array",
                                  "shape": "object"
                                }
                              }
                            }
                          }
                        }
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

03 GET /parameters

RULES []

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {},
    "responses": {
      "200": {
        "schema": {
          "item
```

```
s": {"additional_properties": false, "fields": {"LCL": {"nullable": true, "required": false, "schema": {"type": "number"}}, "LSL": {"nullable": true, "required": false, "schema": {"type": "number"}}, "TARGET": {"nullable": false, "required": true, "schema": {"type": "number"}}, "UCL": {"nullable": false, "required": true, "schema": {"type": "number"}}, "USL": {"nullable": false, "required": true, "schema": {"type": "number"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "ctrl_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "ctrl_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "parameter_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "parameter_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "spec_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "spec_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "target_value": {"nullable": false, "required": true, "schema": {"type": "number"}}, "unit": {"nullable": true, "required": false, "schema": {"type": "string"}}, "upper_only": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object", "type": "array", "shape": "array"}}
04 POST /documents/search
```

RULES []

```
{ "request": { "body": { "additional_properties": false, "fields": { "model_code": { "nullable": true, "required": false, "schema": { "min_length": 1, "type": "string" } }, "query": { "nullable": false, "required": true, "schema": { "max_length": 1000, "min_length": 1, "type": "string" } }, "top_k": { "nullable": false, "required": false, "schema": { "default": 4, "maximum": 10.0, "minimum": 1.0, "type": "integer" } }, "type": "object" }, "header": {}, "path": {}, "query": {}, "responses": { "200": { "schema": { "items": { "additional_properties": false, "fields": { "chunk_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "content": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "doc_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "document_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "model_code": { "nullable": true, "required": false, "schema": { "min_length": 1, "type": "string" } }, "score": { "nullable": false, "required": true, "schema": { "maximum": 1.0, "minimum": -1.0, "type": "number" } }, "section": { "nullable": true, "required": false, "schema": { "type": "string" } }, "title": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "type": "object", "type": "array" }, "shape": "array" }, "422": { "schema": { "additional_properties": true, "fields": { "detail": { "nullable": false, "required": false, "schema": { "items": { "additional_properties": true, "fields": { "loc": { "nullable": false, "required": true, "schema": { "items": { "type": "union", "variants": [{ "type": "string" }, { "type": "integer" } ] }, "type": "array" }, "msg": { "nullable": false, "required": true, "schema": { "type": "string" } }, "type": { "nullable": false, "required": true, "schema": { "type": "string" } }, "type": "object" }, "type": "array" } }, "type": "object", "shape": "object" } }
```

05 GET /relations/chambers/{chamber_id}

```
RULES ["node_count equals len(nodes)", "relationship_count equals len(relationships)"]
{ "request": { "body": null, "header": {}, "path": { "chamber_id": { "nullable": false, "required": true, "schema": { "type": "string" } }, "query": {}, "responses": { "200": { "schema": { "additional_properties": false, "fields": { "graph_revision": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "nodes": { "nullable": false, "required": true, "schema": { "items": { "additional_properties": false, "fields": { "business_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "display_name": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "label": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "properties": { "nullable": false, "required": true, "schema": { "additional_properties": true, "fields": {}, "type": "object" } }, "type": "object", "type": "array" }, "relationships": { "nullable": false, "required": true, "schema": { "items": { "additional_properties": false, "fields": { "id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "source": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "target": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "type": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "type": "object", "type": "array" }, "root_node_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "type": "object", "shape": "object", "422": { "schema": { "additional_properties": true, "fields": { "detail": { "nullable": false, "required": false, "schema": { "items": { "additional_properties": true, "fields": { "loc": { "nullable": false, "required": true, "schema": { "items": { "type": "union", "variants": [{ "type": "string" }, { "type": "integer" } ] }, "type": "array" }, "msg": { "nullable": false, "required": true, "schema": { "type": "string" } }, "type": { "nullable": false, "required": true, "schema": { "type": "string" } }, "type": "object", "type": "array" } }, "type": "object", "shape": "object" } }
```

06 GET /agent/runs

RULES []

```
{ "request": { "body": null, "header": {}, "path": {}, "query": { "date_from": { "nullable": true, "required": false, "schema": { "format": "date", "type": "string" } }, "date_to": { "nullable": true, "required": false, "schema": { "format": "date", "type": "string" } }, "predicted_fault_code": { "nullable": true, "required": false, "schema": { "enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string" } }, "status": { "nullable": true, "required": false, "schema": { "enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string" } }, "responses": { "200": { "schema": { "items": { "additional_properties": false, "fields": { "action_id": { "nullable": true, "required": true, "schema": { "min_length": 1, "type": "string" } }, "agent_run_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "alarm_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "alarm_source": { "nullable": false, "required": true, "schema": { "enum": ["R03", "SUMMARY", "TRACE"], "type": "string" } }, "approval_id": { "nullable": true, "required": true, "schema": { "min_length": 1, "type": "string" } }, "chamber": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "chamber_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "confidence": { "nu
```

llable": true, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object"}, "type": "array"}}, "fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "fault_color": {"nullable": false, "required": true, "schema": {"type": "null"}}, "fault_name": {"nullable": false, "required": true, "schema": {"type": "null"}}, "latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "llm_model": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "predicted_fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "recommended_action": {"nullable": true, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "tools": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"n": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "s": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "type": "array"}, "shape": "array", "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}}

07 POST /agent/ask

RULES [{"fault_code and ground_truth_fault_code are forbidden", "metrology_alarm_result is forbidden"} {"request": {"body": {"additional_properties": false, "fields": {"agent_run_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "question": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object"}, "header": {"path": {}, "query": {}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"answer": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "confidence": {"nullable": true, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "evidence": {"nullable": true, "required": true, "schema": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "doc_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}}, "type": "object"}, "evidence_items": {"nullable": false, "required": true, "schema": {"items": {"discriminator": "type", "type": "discriminated_union", "variants": {"AGENT_RUN": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"max_length": 4000, "min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "ALARM": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "DOCUMENT": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "GRAPH": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "graph_revision": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "relation_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "METROLOGY": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "TRACE": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "limit": {"nullable": false, "required": true, "schema": {"type": "string"}}, "limitations": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "predicted_fault_code": {"nullable": true, "required": true, "schema": {"enum":

```
[{"FOC": "MFD", "OTH": "RFM", "TMD": "TMD", "type": "string"}, {"recommended_action": {"nullable": true, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, {"title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"tools": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, {"type": "object"}, {"shape": "object"}, {"404": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, {"shape": "object"}, {"422": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, {"shape": "object"}]}
```

```
: true, "required": true, "schema": {"type": "string"}}, "decision_comment": {"nullable": true, "required": true, "schema": {"type": "string"}}, "equipment": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "predicted_fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}}
```

10 GET /audit-logs

RULES []

```
{"request": {"body": null, "header": {}, "path": {}, "query": {"actor_type": {"nullable": true, "required": false, "schema": {"type": "string"}}, "date_from": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "date_to": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "entity_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "entity_type": {"nullable": true, "required": false, "schema": {"type": "string"}}, "event_type": {"nullable": true, "required": false, "schema": {"type": "string"}}, "responses": {"200": {"schema": {"items": {"additional_properties": true, "fields": {"actor": {"nullable": false, "required": true, "schema": {"type": "string"}}, "actor_id": {"nullable": true, "required": true, "schema": {"type": "string"}}, "actor_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "after": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "audit_id": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "before": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "detail": {"nullable": true, "required": true, "schema": {"type": "string"}}, "entity": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "type": "object"}, "type": "array", "shape": "array"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}}
```

11 GET /dataset/bounds

RULES []

deferred inventory

12 GET /dashboard/summary

RULES []

```
{"request": {"body": null, "header": {}, "path": {}, "query": {"area": {"nullable": true, "required": false, "schema": {"enum": ["ALL", "Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "date": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "equipment_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "area": {"nullable": true, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "daily_trend": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"date": {"nullable": false, "required": true, "schema": {"format": "date", "type": "string"}}, "has_r03_consec": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "ooc_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "oos_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}, "type": "array"}}, "date_range": {"nullable": false, "required": true, "schema": {"items": {"format": "date", "type": "string"}, "type": "array"}}, "equipment_counts": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chambers": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "array"}}, "hierarchy": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chambers": {"nullable": false, "required": true, "schema": {"items": {"min_length": 1, "type": "string"}, "type": "array"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "ooc_count": {"nullable": false
```

```
, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "oos_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "pending_approvals": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "approval_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "equipment_id": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "request_id_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}, "severity": {"nullable": false, "required": true, "schema": {"enum": ["HIGH", "LOW", "MEDIUM"], "type": "string"}, "type": "object"}, "recent_alarms": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"action_code": {"nullable": true, "required": false, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}, "action_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}, "agent_run_status": {"nullable": true, "required": false, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "approval_status": {"nullable": true, "required": false, "schema": {"enum": ["APPROVED", "AUTO", "EXPIRED", "PENDING", "REJECTED"], "type": "string"}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "detail": {"nullable": false, "required": true, "schema": {"type": "string"}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "hit_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}, "incident": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "judgement": {"nullable": false, "required": true, "schema": {"enum": ["OOC", "OOS"], "type": "string"}, "latest_agent_run_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}, "lot_hist_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}, "recipe_step_name": {"nullable": true, "required": false, "schema": {"type": "string"}, "recipe_step_no": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}, "rule_id": {"nullable": false, "required": true, "schema": {"enum": ["R01_OOS", "R02_OOC", "R03_CONSEC"], "type": "string"}, "sensor_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}, "wafer_no": {"nullable": true, "required": false, "schema": {"minimum": 1.0, "type": "integer"}, "type": "object"}, "type": "array"}, "reference_date": {"nullable": false, "required": true, "schema": {"format": "date", "type": "string"}, "sensor_catalog": {"nullable": false, "required": true, "schema": {"items": {"min_length": 1, "type": "string"}, "type": "array"}, "top_sensors": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}, "chamber_ids": {"nullable": false, "required": true, "schema": {"items": {"min_length": 1, "type": "string"}, "type": "array"}, "sensor_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "type": "object"}, "type": "array"}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}, "type": "object"}, "shape": "object"}}, "13 GET /alarms/{source}/{alarm_id}
RULES []
{"request": {"body": null, "header": {}, "path": {"alarm_id": {"nullable": false, "required": true, "schema": {"type": "string"}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"action_code": {"nullable": true, "required": false, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}, "action_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}, "agent_run_status": {"nullable": true, "required": false, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "approval_status": {"nullable": true, "required": false, "schema": {"enum": ["APPROVED", "AUTO", "EXPIRED", "PENDING", "REJECTED"], "type": "string"}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "detail": {"nullable": false, "required": true, "schema": {"type": "string"}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "hit_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}, "incident": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "judgement": {"nullable": false, "required": true, "schema": {"enum": ["OOC", "OOS"], "type": "string"}, "latest_agent_run_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}, "lot_h
```



```
: {"items": {"type": "integer"}, "type": "array"}}, {"type": "object"}, {"type": "array"}}, {"recipes": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "recipe_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "sensors": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"ctrl_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "ctrl_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "sensor_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "sensor_name": {"nullable": false, "required": true, "schema": {"type": "string"}}, "spec_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "spec_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "target": {"nullable": true, "required": false, "schema": {"type": "number"}}, "unit": {"nullable": true, "required": false, "schema": {"type": "string"}}, "upper_only": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object"}, "type": "array"}}, {"type": "object"}, {"shape": "object"}}}
16 POST /traces/search
RULES []
{"request": {"body": {"additional_properties": false, "fields": {"area": {"nullable": true, "required": false, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "from": {"nullable": true, "required": false, "schema": {"format": "date-time", "type": "string"}}, "lot_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "recipe_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "sensor_ids": {"nullable": true, "required": false, "schema": {"items": {"min_length": 1, "type": "string"}, "type": "array"}}, "to": {"nullable": true, "required": false, "schema": {"format": "date-time", "type": "string"}}, "wafer_nos": {"nullable": true, "required": false, "schema": {"items": {"type": "integer"}, "type": "array"}}, "type": "object"}, "header": {}, "path": {}, "query": {}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"limits": {"nullable": false, "required": true, "schema": {"additional_properties": {"additional_properties": false, "fields": {"ctrl_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "ctrl_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "sensor_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "sensor_name": {"nullable": false, "required": true, "schema": {"type": "string"}}, "spec_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "spec_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "target": {"nullable": true, "required": false, "schema": {"type": "number"}}, "unit": {"nullable": true, "required": false, "schema": {"type": "string"}}, "upper_only": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object"}, "fields": {}, "type": "object"}, "measured_step_stats": {"nullable": false, "required": false, "schema": {"additional_properties": {"additional_properties": true, "fields": {}, "type": "object"}, "fields": {}, "type": "object"}, "total": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "wafers": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_hist_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "missing_steps": {"nullable": false, "required": false, "schema": {"items": {"type": "string"}, "type": "array"}}, "occurred_at": {"nullable": true, "required": false, "schema": {"format": "date-time", "type": "string"}}, "points": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"measured_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "recipe_step_name": {"nullable": true, "required": false, "schema": {"type": "string"}}, "recipe_step_no": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "seq_no": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "value": {"nullable": false, "required": true, "schema": {"type": "number"}}, "type": "object"}, "type": "array"}, "sensor_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "wafer_no": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "type": "object"}, "type": "array"}}, {"type": "object"}, {"shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}}, {"type": "object"}, {"shape": "object"}}}
17 GET /relations/equipment/{equipment_id}
RULES []
deferred inventory
18 GET /documents/{document_id}
RULES []
{"request": {"body": null, "header": {}, "path": {"document_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "query": {}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"chunks": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chunk_seq": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "content": {"nullable": false, "required": true, "schema": {"type": "string"}}, "section_title": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, {"doc_type": {"nullable": true, "required": false, "schema": {"enum": ["MANUAL", "SPEC", "TROUBLESHOOT"], "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1,
```



```
"type": "string"}}, "model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "source_path": {"nullable": true, "required": false, "schema": {"type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"type": "string"}}, "version": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object", "shape": "object", "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}}, "type": "object", "type": "array"}}, "type": "object", "shape": "object"}}
```

19 POST /agent/runs

RULES []

```
{request: {"body": {"additional_properties": false, "fields": {"alarm": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}}, "type": "object"}, "header": {}, "path": {}, "query": {}, "responses": {"202": {"schema": {"additional_properties": false, "fields": {"agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object", "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object", "type": "array"}}, "type": "object", "shape": "object"}}
```

20 GET /agent/runs/{run_id}

RULES []

```
{request: {"body": null, "header": {}, "path": {"run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"action": {"nullable": true, "required": true, "schema": {"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_status": {"nullable": true, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "completed_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "started_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object", "type": "array"}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "action_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "approval": {"nullable": true, "required": true, "schema": {"additional_properties": false, "fields": {"action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "decided_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "decision_comment": {"nullable": true, "required": true, "schema": {"type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "type": "object"}, "approval_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "confidence": {"nullable": true, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object", "type": "array"}, "diagnosis": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"alternative_hypotheses": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "summary": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object", "type": "array"}, "cause_summary": {"nullable": true, "required": false, "schema": {"max_length": 2000, "type": "string"}}, "confidence": {"nullable": true, "required": false, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "diagnostic_coverage": {"nullable": true, "required": false, "schema": {"type": "string"}}, "evidence_synthesis": {"nullable": true, "required": false, "schema": {"type": "string"}}, "evidence_synthesis": {"nullable": true, "required": false, "schema": {"type": "string"}}
```

[illegible]

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {},
    "responses": {
      "200": {
        "schema": {
          "additional_properties": false,
          "fields": {
            "fault_5class": {
              "nullable": true,
              "required": true,
              "schema": {
                "additional_properties": false,
                "fields": {
                  "classification": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "additional_properties": false,
                      "fields": {
                        "accuracy": {
                          "nullable": false,
                          "required": true,
                          "schema": {
                            "additional_properties": false,
                            "fields": {
                              "denominator": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "minimum": 0.0,
                                  "type": "integer"
                                },
                                "numerator": {
                                  "nullable": false,
                                  "required": true,
                                  "schema": {
                                    "minimum": 0.0,
                                    "type": "integer"
                                  },
                                  "rate": {
                                    "nullable": false,
                                    "required": true,
                                    "schema": {
                                      "maximum": 1.0,
                                      "minimum": 0.0,
                                      "type": "number"
                                    }
                                  }
                                },
                                "type": "object"
                              },
                              "by_class": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "additional_properties": {
                                    "additional_properties": false,
                                    "fields": {
                                      "f1": {
                                        "nullable": false,
                                        "required": true,
                                        "schema": {
                                          "maximum": 1.0,
                                          "minimum": 0.0,
                                          "type": "number"
                                        },
                                        "false_negative": {
                                          "nullable": false,
                                          "required": true,
                                          "schema": {
                                            "minimum": 0.0,
                                            "type": "integer"
                                          },
                                          "false_positive": {
                                            "minimum": 0.0,
                                            "type": "integer"
                                          }
                                        }
                                      }
                                    }
                                  }
                                }
                              }
                            }
                          }
                        }
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

```
nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "precision": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "recall": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "support": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "true_positive": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}, {"fields": {}, "type": "object"}}, {"macro_f1_5class": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, {"observed_class_macro_f1": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, {"population_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, {"unclassified_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, {"type": "object"}, {"evidence_valid_run": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"denominator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, {"numerator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, {"rate": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "type": "object"}}, {"exclusions": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, {"meaning": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"reason": {"nullable": true, "schema": {"enum": ["AMBIGUOUS_LABEL", "NO_INJECTED_FAULT"], "type": "string"}}, "type": "object"}, "type": "array"}}, {"hard_gate_passed": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, {"hard_gate_reasons": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, {"label_source": {"nullable": false, "required": true, "schema": {"type": "string"}}, {"metrology_observed_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, {"metrology_total_lot_hist_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, {"production_ground_truth_available": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, {"production_performance_disclaimer": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"public_fault_ground_truth_available": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, {"rule_action_agreement": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"denominator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, {"numerator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, {"rate": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "type": "object"}}, {"structured_prediction": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"denominator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, {"numerator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, {"rate": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "type": "object"}}, {"usage_scope": {"nullable": false, "required": true, "schema": {"type": "string"}}, {"versions": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"dataset_epoch": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"model_version": {"nullable": true, "required": true, "schema": {"type": "string"}}, {"policy_version": {"nullable": true, "required": true, "schema": {"type": "string"}}, {"prompt_version": {"nullable": true, "required": true, "schema": {"type": "string"}}, "type": "object"}}, {"type": "object"}}, {"fault_5class_empty_reason": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_VIOLATION", "NOT_CONFIGURED", "NOT_FOUND"], "type": "string"}}, {"golden_flow": {"nullable": true, "required": true, "schema": {"additional_properties": false, "fields": {"dataset_epoch": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"phases": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"metrics": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, {"phase": {"nullable": false, "required": true, "schema": {"enum": ["BATCH_BASELINE", "DECISIONS", "MANUAL_RETRY", "PRE_FLIGHT", "PRE_APPROVAL", "SECOND_BATCH", "UNKNOWN"], "type": "string"}}, {"reasons": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, {"status": {"nullable": false, "required": true, "schema": {"enum": ["EVIDENCE_INCOMPLETE", "FAIL", "PASS"], "type": "string"}}, "type": "object"}, {"type": "array"}}, {"status": {"nullable": false, "required": true, "schema": {"enum": ["EVIDENCE_INCOMPLETE", "FAIL", "PASS"], "type": "string"}}, "type": "object"}}, {"golden_flow_empty_reason": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_VIOLATION", "NOT_CONFIGURED", "NOT_FOUND"], "type": "string"}}, "type": "object"}, {"shape": "object"}}}
```

22 POST /agent/runs/{run_id}/retry

RULES []

deferred inventory

23 GET /agent/runs/paged

RULES []

deferred inventory

24 GET /approvals/paged

RULES []

deferred inventory

25 GET /actions

RULES []

```
{ "request": { "body": null, "header": {}, "path": {}, "query": { "action_code": { "nullable": true, "required": false, "schema": { "enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string" } } }, "responses": { "200": { "schema": { "items": { "additional_properties": false, "fields": { "action_code": { "nullable": false, "required": true, "schema": { "enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string" } }, "action_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "agent_run_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "approval_status": { "nullable": true, "required": true, "schema": { "enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string" } }, "chamber": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }
```

```
{ "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "created_by_agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELLED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object"}, "type": "array"}}, "equipment": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}, "shape": "array", "422": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}}}

26 GET /actions/{action_id}
RULES []

{"request": {"body": null, "header": {}, "path": {"action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_status": {"nullable": true, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "created_by_agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "completed_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object"}, "type": "array"}}, "equipment": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "shape": "object"}, "404": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}, "503": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}}}}

27 POST /actions/{action_id}/deliveries/{channel}/retry
RULES []

deferred inventory

28 POST /internal/actions/{action_id}/delivery
```

```
RULES [{"FAILED": "requires error_code non-null and permits provider_message_id null", "SENT": "requires provide\nr_message_id non-null and error_code null"}]\n{"request": {"body": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required\n": true, "schema": {"enum": ["EMAIL", "MES MOCK"], "type": "string"}}, "completed_at": {"nullable": false\n, "required": true, "schema": {"format": "date-time", "type": "string"}}, "error_code": {"nullable": true\n, "required": true, "schema": {"min_length": 1, "type": "string"}}, "event_id": {"nullable": false, "requ\nired": true, "schema": {"min_length": 1, "type": "string"}}, "provider_message_id": {"nullable": true, "r\nquired": true, "schema": {"min_length": 1, "type": "string"}}, "request_hash": {"nullable": false, "requ\nired": true, "schema": {"max_length": 64, "min_length": 64, "pattern": "\\[0-9a-f]{64}$", "type": "string\n"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAILED", "SENT"], "type": "strin\ng"}}, "type": "object"}, "header": {"X-Delivery-Signature": {"nullable": true, "required": false, "schem\na": {"type": "string"}}, "X-Delivery-Timestamp": {"nullable": true, "required": false, "schema": {"type":\n"string"}}, "path": {"action_id": {"nullable": false, "required": true, "schema": {"type": "string"}},\n"query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"action_id": {\n"nullable": false, "required": true, "schema": {"max_length": 20, "min_length": 1, "type": "string"}}, "c\nhannel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES MOCK"], "type": "string\n"}}, "completed_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "stri\nng"}}, "duplicate": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "error_code": {\n"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "provider_message_id"\n: {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "request_hash": {\nnullable": false, "required": true, "schema": {"max_length": 64, "min_length": 64, "pattern": "\\[0-9a-f]{\n64}$", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAILED",\n"SENT"], "type": "string"}}, "type": "object"}, "shape": "object", "422": {"schema": {"additional_prope\nrties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additiona\nl_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type":\n"union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable":\nfalse, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "sc\nhema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}\n}}\n29 POST /analytics/query\nRULES []\n{"request": {"body": {"additional_properties": false, "fields": {"question": {"nullable": false, "require\nd": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object"}, "header\n": {"path": {}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields\n": {"columns": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "arr\nay"}}, "cross_check": {"nullable": true, "required": false, "schema": {"additional_properties": false, "f\nields": {"cypher": {"nullable": true, "required": false, "schema": {"type": "string"}}, "status": {"nulla\nble": false, "required": true, "schema": {"enum": ["MATCH", "MISMATCH", "SKIPPED"], "type": "string"}}, "\nsummary": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}, "erro\nr_msg": {"nullable": true, "required": false, "schema": {"type": "string"}}, "generated_sql": {"nullable\n": true, "required": false, "schema": {"type": "string"}}, "group_by": {"nullable": false, "required": tru\ne, "schema": {"items": {"type": "string"}, "type": "array"}}, "is_rejected": {"nullable": false, "require\nd": true, "schema": {"type": "boolean"}}, "is_valid": {"nullable": false, "required": true, "schema": {"t\nype": "boolean"}}, "latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type":\n"integer"}}, "metric": {"nullable": true, "required": false, "schema": {"additional_properties": false,\n"fields": {"column": {"nullable": true, "required": false, "schema": {"type": "string"}}, "p": {"nullabl\n": true, "required": false, "schema": {"maximum": 100.0, "minimum": 0.0, "type": "number"}}, "type": {"nu\nllable": false, "required": true, "schema": {"enum": ["count", "max", "mean", "median", "min", "percentil\n", "ratio", "std", "sum"], "type": "string"}}, "type": "object"}, "metric_result": {"nullable": false,\n"required": false, "schema": {"type": "union", "variants": [{"type": "integer"}, {"type": "number"}, {"i\ntems": {"additional_properties": false, "fields": {"group": {"nullable": false, "required": true, "schema\n": {"additional_properties": true, "fields": {}, "type": "object"}}, "value": {"nullable": false, "requir\ned": false, "schema": {"type": "union", "variants": [{"type": "integer"}, {"type": "number"}, {"type": "n\null"}]}}, "type": "object"}, "type": "array"}, {"type": "null"}]}, "nl_query_log_id": {"nullable": true\n, "required": false, "schema": {"minimum": 1.0, "type": "integer"}}, "question": {"nullable": false, "req\nquired": true, "schema": {"type": "string"}}, "reject_reason": {"nullable": true, "required": false, "sche\nma": {"type": "string"}}, "row_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "\ntype": "integer"}}, "rows": {"nullable": false, "required": true, "schema": {"items": {"additional_proper\nities": true, "fields": {}, "type": "object"}, "type": "array"}}, "visualization": {"nullable": true, "req\nquired": false, "schema": {"additional_properties": false, "fields": {"chart_type": {"nullable": false, "r\nquired": true, "schema": {"enum": ["bar", "histogram", "line", "table"], "type": "string"}}, "x": {"null\nable": true, "required": false, "schema": {"type": "string"}}, "y": {"nullable": true, "required": false,\n"schema": {"type": "string"}}, "type": "object"}}, "type": "object"}, "shape": "object", "422": {"sch\nema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema\n": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "sch\nema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array\n"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": fals\n", "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "obje\nct"}, "shape": "object"}]}}
```

29 POST /analytics/query

RULES []

```
{"request": {"body": {"additional_properties": false, "fields": {"question": {"nullable": false, "require\nd": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object"}, "header\n": {"path": {}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields\n": {"columns": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "arr\nay"}}, "cross_check": {"nullable": true, "required": false, "schema": {"additional_properties": false, "f\nields": {"cypher": {"nullable": true, "required": false, "schema": {"type": "string"}}, "status": {"nulla\nble": false, "required": true, "schema": {"enum": ["MATCH", "MISMATCH", "SKIPPED"], "type": "string"}}, "\nsummary": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}, "erro\nr_msg": {"nullable": true, "required": false, "schema": {"type": "string"}}, "generated_sql": {"nullable\n": true, "required": false, "schema": {"type": "string"}}, "group_by": {"nullable": false, "required": tru\n", "schema": {"items": {"type": "string"}, "type": "array"}}, "is_rejected": {"nullable": false, "require\nd": true, "schema": {"type": "boolean"}}, "is_valid": {"nullable": false, "required": true, "schema": {"t\nype": "boolean"}}, "latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type":\n"integer"}}, "metric": {"nullable": true, "required": false, "schema": {"additional_properties": false,\n"fields": {"column": {"nullable": true, "required": false, "schema": {"type": "string"}}, "p": {"nullabl\n": true, "required": false, "schema": {"maximum": 100.0, "minimum": 0.0, "type": "number"}}, "type": {"nu\nllable": false, "required": true, "schema": {"enum": ["count", "max", "mean", "median", "min", "percentil\n", "ratio", "std", "sum"], "type": "string"}}, "type": "object"}, "metric_result": {"nullable": false,\n"required": false, "schema": {"type": "union", "variants": [{"type": "integer"}, {"type": "number"}, {"i\ntems": {"additional_properties": false, "fields": {"group": {"nullable": false, "required": true, "schema\n": {"additional_properties": true, "fields": {}, "type": "object"}}, "value": {"nullable": false, "requir\ned": false, "schema": {"type": "union", "variants": [{"type": "integer"}, {"type": "number"}, {"type": "n\null"}]}}, "type": "object"}, "type": "array"}, {"type": "null"}]}, "nl_query_log_id": {"nullable": true\n, "required": false, "schema": {"minimum": 1.0, "type": "integer"}}, "question": {"nullable": false, "req\nquired": true, "schema": {"type": "string"}}, "reject_reason": {"nullable": true, "required": false, "sche\nma": {"type": "string"}}, "row_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "\ntype": "integer"}}, "rows": {"nullable": false, "required": true, "schema": {"items": {"additional_proper\nities": true, "fields": {}, "type": "object"}, "type": "array"}}, "visualization": {"nullable": true, "req\nquired": false, "schema": {"additional_properties": false, "fields": {"chart_type": {"nullable": false, "r\nquired": true, "schema": {"enum": ["bar", "histogram", "line", "table"], "type": "string"}}, "x": {"null\nable": true, "required": false, "schema": {"type": "string"}}, "y": {"nullable": true, "required": false,\n"schema": {"type": "string"}}, "type": "object"}}, "type": "object"}, "shape": "object", "422": {"sch\nema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema\n": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "sch\nema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array\n"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": fals\n", "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "obje\nct"}, "shape": "object"}]}}
```

30 POST /analytics/graph-query

RULES []

```
{"request": {"body": {"additional_properties": false, "fields": {"question": {"nullable": false, "require\nd": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object"}, "header\n": {"path": {}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields
```

```
": {"columns": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}, "error_msg": {"nullable": true, "required": true, "schema": {"type": "string"}, "generated_cypher": {"nullable": true, "required": false, "schema": {"type": "string"}, "is_rejected": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "is_valid": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "question": {"nullable": false, "required": true, "schema": {"type": "string"}, "reject_reason": {"nullable": true, "required": false, "schema": {"type": "string"}, "row_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}, "rows": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": true, "fields": {}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}}}
```

31 POST /analytics/validate

RULES []

```
{"request": {"body": {"additional_properties": false, "fields": {"sql": {"nullable": false, "required": true, "schema": {"max_length": 20000, "min_length": 1, "type": "string"}}, "type": "object"}, "header": {}, "path": {}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"checks": {"nullable": true, "required": false, "schema": {"items": {"additional_properties": false, "fields": {"key": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "label": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "ok": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object"}, "type": "array"}, "normalized_sql": {"nullable": true, "required": false, "schema": {"type": "string"}, "reason": {"nullable": false, "required": true, "schema": {"type": "string"}, "valid": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}}}
```

32 GET /analytics/history

RULES []

```
{"request": {"body": null, "header": {}, "path": {}, "query": {"date_from": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}, "date_to": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}, "is_rejected": {"nullable": true, "required": false, "schema": {"type": "boolean"}}, "is_valid": {"nullable": true, "required": false, "schema": {"type": "boolean"}}, "page": {"nullable": false, "required": false, "schema": {"default": 1, "minimum": 1, "type": "integer"}, "size": {"nullable": false, "required": false, "schema": {"default": 20, "maximum": 100, "minimum": 1, "type": "integer"}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"asked_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}, "error_msg": {"nullable": true, "required": false, "schema": {"type": "string"}, "generated_sql": {"nullable": true, "required": false, "schema": {"type": "string"}, "is_rejected": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "is_valid": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "latency_ms": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}, "nl_query_log_id": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}, "outcome": {"nullable": false, "required": true, "schema": {"enum": ["DB_ERROR", "POLICY_REJECTED", "SUCCESS", "VALIDATION_FAILED"], "type": "string"}, "question": {"nullable": false, "required": true, "schema": {"type": "string"}, "reject_reason": {"nullable": true, "required": false, "schema": {"type": "string"}, "row_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}, "type": "array"}, "page": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}, "size": {"nullable": false, "required": true, "schema": {"maximum": 100.0, "minimum": 1.0, "type": "integer"}, "total": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}}}
```

33 GET /analytics/evaluations

RULES []

```
{"request": {"body": null, "header": {}, "path": {}, "query": {"latest": {"nullable": false, "required": false, "schema": {"default": true, "type": "boolean"}, "page": {"nullable": false, "required": false, "schema": {"default": 1, "minimum": 1, "type": "integer"}, "size": {"nullable": false, "required": false, "schema": {"default": 20, "maximum": 100, "minimum": 1, "type": "integer"}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"accuracy": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}, "correct": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}, "defense_passed": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}, "defense_total": {"nullable": false, "required": t
```

```

    rue, "schema": {"minimum": 0.0, "type": "integer"}}, "executed_at": {"nullable": false, "required": true,
    "schema": {"format": "date-time", "type": "string"}}, "items": {"nullable": false, "required": true, "sc
    hema": {"items": {"additional_properties": false, "fields": {"actual_result": {"nullable": true, "require
    d": false, "schema": {"type": "any"}}, "actual_visualization": {"nullable": true, "required": false, "sch
    ema": {"additional_properties": false, "fields": {"chart_type": {"nullable": false, "required": true, "sc
    hema": {"enum": ["bar", "histogram", "line", "table"], "type": "string"}}, "x": {"nullable": true, "requi
    red": false, "schema": {"type": "string"}}, "y": {"nullable": true, "required": false, "schema": {"type":
    "string"}}, "type": "object"}}, "attempt_count": {"nullable": false, "required": true, "schema": {"mini
    mum": 0.0, "type": "integer"}}, "case_id": {"nullable": false, "required": true, "schema": {"min_length":
    1, "type": "string"}}, "case_type": {"nullable": false, "required": true, "schema": {"enum": ["DEFENSE",
    "GOLD"], "type": "string"}}, "expected_result": {"nullable": true, "required": false, "schema": {"type":
    "any"}}, "expected_visualization": {"nullable": true, "required": false, "schema": {"additional_properti
    es": false, "fields": {"chart_type": {"nullable": false, "required": true, "schema": {"enum": ["bar", "hi
    stogram", "line", "table"], "type": "string"}}, "x": {"nullable": true, "required": false, "schema": {"ty
    pe": "string"}}, "y": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "obje
    ct"}}, "generated_sql": {"nullable": true, "required": false, "schema": {"type": "string"}}, "latency_ms"
    : {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "passed": {"nulla
    ble": false, "required": true, "schema": {"type": "boolean"}}, "question": {"nullable": true, "required":
    false, "schema": {"type": "string"}}, "reason": {"nullable": true, "required": false, "schema": {"type":
    "string"}}, "type": "object"}, "type": "array"}, "model": {"nullable": false, "required": true, "schem
    a": {"min_length": 1, "type": "string"}}, "prompt_version": {"nullable": false, "required": true, "schema
    ": {"min_length": 1, "type": "string"}}, "provider": {"nullable": false, "required": true, "schema": {"mi
    n_length": 1, "type": "string"}}, "run_id": {"nullable": false, "required": true, "schema": {"min_length"
    : 1, "type": "string"}}, "temperature": {"nullable": false, "required": true, "schema": {"type": "number"
    }}, "total": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type
    ": {"object"}, "type": "array"}, "page": {"nullable": false, "required": true, "schema": {"minimum": 1.0,
    "type": "integer"}}, "size": {"nullable": false, "required": true, "schema": {"maximum": 100.0, "minimum
    ": 1.0, "type": "integer"}}, "total": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "t
    ype": "integer"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": tr
    ue, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properti
    es": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union",
    "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "re
    quired": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"t
    ype": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}}}
```

34 GET /audit-logs/paged

RULES []

```

{"request": {"body": null, "header": {}, "path": {}, "query": {"actor_type": {"nullable": true, "required
": false, "schema": {"type": "string"}}, "date_from": {"nullable": true, "required": false, "schema": {"f
ormat": "date", "type": "string"}}, "date_to": {"nullable": true, "required": false, "schema": {"format":
"date", "type": "string"}}, "entity_id": {"nullable": true, "required": false, "schema": {"type": "strin
g"}}, "entity_type": {"nullable": true, "required": false, "schema": {"type": "string"}}, "event_type": {
"nullable": true, "required": false, "schema": {"type": "string"}}, "page": {"nullable": false, "required
": false, "schema": {"default": 1, "minimum": 1, "type": "integer"}}, "size": {"nullable": false, "requir
ed": false, "schema": {"default": 20, "maximum": 100, "minimum": 1, "type": "integer"}}, "responses": {
"200": {"schema": {"additional_properties": true, "fields": {"event_type_counts": {"nullable": false, "re
quired": true, "schema": {"additional_properties": {"type": "integer"}, "fields": {}, "type": "object"}},
"event_types": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "ar
ray"}}, "items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": true
, "fields": {"actor": {"nullable": false, "required": true, "schema": {"type": "string"}}, "actor_id": {"
nullable": true, "required": true, "schema": {"type": "string"}}, "actor_type": {"nullable": false, "requ
ired": true, "schema": {"type": "string"}}, "after": {"nullable": true, "required": true, "schema": {"add
itional_properties": true, "fields": {}, "type": "object"}}, "at": {"nullable": false, "required": true,
"schema": {"format": "date-time", "type": "string"}}, "audit_id": {"nullable": false, "required": true, "
schema": {"type": "integer"}}, "before": {"nullable": true, "required": true, "schema": {"additional_prop
erties": true, "fields": {}, "type": "object"}}, "detail": {"nullable": true, "required": true, "schema":
{"type": "string"}}, "entity": {"nullable": false, "required": true, "schema": {"type": "string"}}, "ent
ity_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_type": {"nullable":
false, "required": true, "schema": {"type": "string"}}, "event": {"nullable": false, "required": true, "
schema": {"type": "string"}}, "event_type": {"nullable": false, "required": true, "schema": {"type": "str
ing"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "s
tring"}}, "type": "object"}, "type": "array"}, "page": {"nullable": false, "required": true, "schema":
{"type": "integer"}}, "size": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "tota
l": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "type": "object"}, "shape": "o
bject"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "requ
ired": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "
required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"
}]}}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "typ
e": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "arr
ay"}}, "type": "object"}, "shape": "object"}}}
```

35 GET /health

RULES []

```

{"request": {"body": null, "header": {}, "path": {}, "query": {}}, "responses": {"200": {"schema": {"addi
tional_properties": false, "fields": {"status": {"nullable": false, "required": true, "schema": {"type":
"string"}}, "type": "object"}, "shape": "object"}}
```


[illegible]

DTO COMPONENTS

ActionCode

```
{ "enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string" }
```

ActionDeliveryDetailItem

```
{ "additional_properties": false, "fields": { "channel": { "nullable": false, "required": true, "schema": { "enum": ["EMAIL", "MES"], "type": "string" }, "completed_at": { "nullable": true, "required": true, "schema": { "format": "date-time", "type": "string" }, "started_at": { "nullable": true, "required": true, "schema": { "format": "date-time", "type": "string" }, "status": { "nullable": false, "required": true, "schema": { "enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string" }, "type": "object" }
```

ActionDeliveryItem

```
{ "additional_properties": false, "fields": { "channel": { "nullable": false, "required": true, "schema": { "enum": ["EMAIL", "MES"], "type": "string" }, "status": { "nullable": false, "required": true, "schema": { "enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string" }, "type": "object" }
```

ActionDetailResponse

```
{ "additional_properties": false, "fields": { "action_code": { "nullable": false, "required": true, "schema": { "enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string" }, "action_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "agent_run_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "approval_status": { "nullable": true, "required": true, "schema": { "enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string" }, "chamber": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "chamber_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "created_at": { "nullable": false, "required": true, "schema": { "format": "date-time", "type": "string" }, "created_by_agent_run_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "deliveries": { "nullable": false, "required": true, "schema": { "items": { "additional_properties": false, "fields": { "channel": { "nullable": false, "required": true, "schema": { "enum": ["EMAIL", "MES"], "type": "string" }, "completed_at": { "nullable": true, "required": true, "schema": { "format": "date-time", "type": "string" }, "started_at": { "nullable": true, "required": true, "schema": { "format": "date-time", "type": "string" }, "status": { "nullable": true, "required": true, "schema": { "enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string" }, "type": "object" }, "type": "array" }, "equipment": { "nullable": true, "required": true, "schema": { "min_length": 1, "type": "string" }, "equipment_id": { "nullable": true, "required": true, "schema": { "min_length": 1, "type": "string" }, "lot": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "lot_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "reason": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "type": "object" }
```

ActionItem

```
{ "additional_properties": false, "fields": { "action_code": { "nullable": false, "required": true, "schema": { "enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string" }, "action_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "agent_run_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "approval_status": { "nullable": true, "required": true, "schema": { "enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string" }, "chamber": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "chamber_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "created_at": { "nullable": false, "required": true, "schema": { "format": "date-time", "type": "string" }, "created_by_agent_run_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "deliveries": { "nullable": false, "required": true, "schema": { "items": { "additional_properties": false, "fields": { "channel": { "nullable": false, "required": true, "schema": { "enum": ["EMAIL", "MES"], "type": "string" }, "status": { "nullable": false, "required": true, "schema": { "enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string" }, "type": "object" }, "type": "array" }, "equipment": { "nullable": true, "required": true, "schema": { "min_length": 1, "type": "string" }, "equipment_id": { "nullable": true, "required": true, "schema": { "min_length": 1, "type": "string" }, "lot": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "lot_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "reason": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "type": "object" }
```

AgentAskCompatibilityEvidence

```
{ "additional_properties": false, "fields": { "chunk_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "doc_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "document_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "section": { "nullable": true, "required": true, "schema": { "type": "string" }, "type": "object" }
```

AgentAskRequest

```
{ "additional_properties": false, "fields": { "agent_run_id": { "nullable": true, "required": false, "schema": { "min_length": 1, "type": "string" }, "question": { "nullable": false, "required": true, "schema": { "max_length": 1000, "min_length": 1, "type": "string" }, "type": "object" }
```

AgentAskResponse

```
{ "additional_properties": false, "fields": { "answer": { "nullable": false, "required": true, "schema": { "type": "string" }, "confidence": { "nullable": true, "required": true, "schema": { "maximum": 1, "minimum": 0, "type": "number" }, "evidence": { "nullable": true, "required": true, "schema": { "additional_properties": false, "fields": { "chunk_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "doc_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "document_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "section": { "nullable": true, "required": true, "schema": { "type": "string" }, "type": "object" }, "evidence_items": { "nullable": false, "required": true, "schema": { "items": { "discriminator": "type", "type":
```

```
"discriminated_union", "variants": {"ALARM": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["ALARM"], "type": "string"}}}, "type": "object"}, "DOCUMENT": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["DOCUMENT"], "type": "string"}}}, "type": "object"}, "GRAPH": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "graph_revision": {"nullable": false, "required": true, "schema": {"pattern": "^0-9a-f{64}$", "type": "string"}}, "relation_id": {"nullable": false, "required": true, "schema": {"pattern": "^REL-[0-9a-f]{20}$", "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["GRAPH"], "type": "string"}}, "type": "object"}, "METROLOGY": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["METROLOGY"], "type": "string"}}, "rules": [{"metrology_alarm_result_is forbidden": true, "type": "object"}, "TRACE": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["TRACE"], "type": "string"}}, "type": "object"}], "type": "array"}, "limit": {"nullable": false, "required": true, "schema": {"type": "string"}}, "limitations": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "predicted_fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "recommended_action": {"nullable": true, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "tools": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}], "rules": [{"fault_code and ground_truth_fault_code are forbidden"], "type": "object"}}
```

AgentEvaluationResponse

```
{ "additional_properties": false, "fields": { "fault_5class": { "nullable": true, "required": true, "schema": { "additional_properties": false, "fields": { "classification": { "nullable": false, "required": true, "schema": { "additional_properties": false, "fields": { "accuracy": { "nullable": false, "required": true, "schema": { "additional_properties": false, "fields": { "denominator": { "nullable": false, "required": true, "schema": { "minimum": 0.0, "type": "integer"}, "numerator": { "nullable": false, "required": true, "schema": { "minimum": 0.0, "type": "integer"}, "rate": { "nullable": false, "required": true, "schema": { "maximum": 1.0, "minimum": 0.0, "type": "number"}}, "type": "object"}, "by_class": { "nullable": false, "required": true, "schema": { "additional_properties": { "additional_properties": false, "fields": { "f1": { "nullable": false, "required": true, "schema": { "maximum": 1.0, "minimum": 0.0, "type": "number"}}, "false_negative": { "nullable": false, "required": true, "schema": { "minimum": 0.0, "type": "integer"}, "false_positive": { "nullable": false, "required": true, "schema": { "minimum": 0.0, "type": "integer"}, "precision": { "nullable": false, "required": true, "schema": { "maximum": 1.0, "minimum": 0.0, "type": "number"}, "recall": { "nullable": false, "required": true, "schema": { "maximum": 1.0, "minimum": 0.0, "type": "number"}, "support": { "nullable": false, "required": true, "schema": { "minimum": 0.0, "type": "integer"}, "true_positive": { "nullable": false, "required": true, "schema": { "minimum": 0.0, "type": "integer"}}, "type": "object"}, "fields": {}, "type": "object"}, "macro_f1_5class": { "nullable": false, "required": true, "schema": { "maximum": 1.0, "minimum": 0.0, "type": "number"}, "observed_class_macro_f1": { "nullable": false, "required": true, "schema": { "maximum": 1.0, "minimum": 0.0, "type": "number"}, "population_count": { "nullable": false, "required": true, "schema": { "minimum": 0.0, "type": "integer"}, "unclassified_count": { "nullable": false, "required": true, "schema": { "minimum": 0.0, "type": "integer"}}, "type": "object"}, "evidence_valid_run": { "nullable": false, "required": true, "schema": { "additional_properties": false, "fields": { "denominator": { "nullable": false, "required": true, "schema": { "minimum": 0.0, "type": "integer"}, "numerator": { "nullable": false, "required": true, "schema": { "minimum": 0.0, "type": "integer"}, "rate": { "nullable": false, "required": true, "schema": { "maximum": 1.0, "minimum": 0.0, "type": "number"}}, "type": "object"}, "exclusions": { "nullable": false, "required": true, "schema": { "items": { "additional_properties": false, "fields": { "count": { "nullable": false, "required": true, "schema": { "minimum": 0.0, "type": "integer"}, "meaning": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string"}, "reason": { "nullable": false, "required": true, "schema": { "enum": ["AMBIGUOUS_LABEL", "NO_INJECTED_FAULT"], "type": "string"}}, "type": "object"}, "type": "array"}, "hard_gate_passed": { "nullable": false, "required": true, "schema": {"type": "boolean"}}, "hard_gate_reasons": { "nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}, "label_source": { "nullable": false, "required": true, "schema": {"type": "string"}}, "metrology_observed_count": { "nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}, "metrology_total_lot_hist_count": {
```

nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "production_ground_truth_available": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "production_performance_disclaimer": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "public_fault_ground_truth_available": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "rule_action_agreement": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"denominator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "numerator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "rate": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}}, "type": "object"}}, "structured_prediction": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"denominator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "numerator": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "rate": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}}, "type": "object"}}, "usage_scope": {"nullable": false, "required": true, "schema": {"type": "string"}}, "versions": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"dataset_epoch": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "model_version": {"nullable": true, "required": true, "schema": {"type": "string"}}, "policy_version": {"nullable": true, "required": true, "schema": {"type": "string"}}, "prompt_version": {"nullable": true, "required": true, "schema": {"type": "string"}}, "type": "object"}}, "type": "object"}}, "fault_5class_empty_reason": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_VIOLATION", "NOT_CONFIGURED", "NOT_FOUND"], "type": "string"}}, "golden_flow": {"nullable": true, "required": true, "schema": {"additional_properties": false, "fields": {"dataset_epoch": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "phases": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"metrics": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "phase": {"nullable": false, "required": true, "schema": {"enum": ["BATCH_BASELINE", "DECISIONS", "MANUAL_RETRY", "PREFLIGHT", "PRE_APPROVAL", "SECOND_BATCH", "UNKNOWN"], "type": "string"}}, "reasons": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["EVIDENCE_INCOMPLETE", "FAIL", "PASS"], "type": "string"}}, "type": "object"}, "type": "array"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["EVIDENCE_INCOMPLETE", "FAIL", "PASS"], "type": "string"}}, "type": "object"}}, "golden_flow_empty_reason": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_VIOLATION", "NOT_CONFIGURED", "NOT_FOUND"], "type": "string"}}, "type": "object"}

AgentPredictionDetailItem

{"additional_properties": false, "fields": {"cause_summary": {"nullable": false, "required": true, "schema": {"max_length": 2000, "min_length": 1, "type": "string"}}, "confidence": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "generated_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "input_tokens": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "llm_model": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "output_tokens": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "predicted_fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "prompt_version": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "supporting_alarms": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "type": "object"}, "type": "array"}}, "supporting_chunk_ids": {"nullable": false, "required": true, "schema": {"items": {"min_length": 1, "type": "string"}, "type": "array"}}, "supporting_relation_ids": {"nullable": false, "required": true, "schema": {"items": {"min_length": 1, "type": "string"}, "type": "array"}}, "uncertainty": {"nullable": false, "required": true, "schema": {"max_length": 1000, "type": "string"}}, "type": "object"}

AgentRunAccepted

{"additional_properties": false, "fields": {"agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "type": "object"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["RUNNING"], "type": "string"}}, "type": "object"}

AgentRunAcceptedResponse

{"additional_properties": false, "fields": {"agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "type": "object"}}, "status": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}

AgentRunActionItem

{"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_status": {"nullable": true, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "completed_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "started_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "st

```
atus": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object"}, "type": "array"}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
AgentRunApprovalItem
{"additional_properties": false, "fields": {"action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "decided_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "decision_comment": {"nullable": true, "required": true, "schema": {"type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "type": "object"}
AgentRunAskEvidence
{"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"max_length": 4000, "min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}
AgentRunCreateRequest
{"additional_properties": false, "fields": {"alarm": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "type": "object"}}, "type": "object"}
AgentRunDetailResponse
{"additional_properties": false, "fields": {"action": {"nullable": true, "required": true, "schema": {"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_status": {"nullable": true, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "complete_d_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "started_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object"}, "type": "array"}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "action_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "approval": {"nullable": true, "required": true, "schema": {"additional_properties": false, "fields": {"action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "decided_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "decision_comment": {"nullable": true, "required": true, "schema": {"type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "type": "object"}}, "approval_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "confidence": {"nullable": true, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object"}, "type": "array"}, "diagnosis": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"lower_rank_reason": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "summary": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}, "cause_summary": {"nullable": true, "required": false, "schema": {"max_length": 2000, "type": "string"}}, "confidence": {"nullable": true, "required": false, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "diagnostic_coverage": {"nullable": true, "required": false, "schema": {"type": "string"}}, "evidence_synthesis": {"nullable": true, "required": false, "schema": {"max_length": 2000, "type": "string"}}, "limitations": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}, "observations": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}, "predicted_fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "reason_code": {"nullable": true, "required": false, "schema": {"pattern": "^[A-Z0-9_]+S$", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["AVAILABLE", "EMPTY"], "type": "string"}}, "verification_steps": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}, "type": "object"}, "evidence_assessment": {"nullable": fa
```

```
"required": true, "schema": {"additional_properties": false, "fields": {"available_sources": {"nulla
ble": false, "required": true, "schema": {"items": {"enum": ["FDC", "GRAPH", "POSTGRES_ROUTE", "RAG"], "t
ype": "string"}, "type": "array"}}, "conflicting_source_ids": {"nullable": false, "required": true, "sche
ma": {"items": {"type": "string"}, "type": "array"}}, "missing_sources": {"nullable": false, "required":
true, "schema": {"items": {"enum": ["FDC", "GRAPH", "POSTGRES_ROUTE", "RAG"], "type": "string"}, "type":
"array"}}, "reason_codes": {"nullable": false, "required": true, "schema": {"items": {"type": "string"},
"type": "array"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["CONFLICT", "EMPTY
", "PARTIAL", "SUFFICIENT"], "type": "string"}}, "type": "object"}}, "evidence_items": {"nullable": fals
e, "required": true, "schema": {"items": {"discriminator": {"type": "type": "discriminated_union", "varian
ts": {"ALARM": {"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true,
"schema": {"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true,
"schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema":
{"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "type": "object"}}, "excerpt": {"nullable": f
alse, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": false,
"required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required"
: true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "sc
hema": {"type": "string"}}, "type": "object"}, "DOCUMENT": {"additional_properties": false, "fields": {"c
hunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "documen
t_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "excerpt": {
"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "section": {"nullabl
e": true, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": t
rue, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "sche
ma": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"typ
e": "string"}, "type": "object"}, "GRAPH": {"additional_properties": false, "fields": {"excerpt": {"null
able": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "graph_revision": {"nulla
ble": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "relation_id": {"nullable"
: false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": fals
e, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "requir
ed": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true,
"schema": {"type": "string"}, "type": "object"}, "METROLOGY": {"additional_properties": false, "fields":
{"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "sour
ce_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"
nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable":
false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "TRACE": {"additional_propert
ies": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "typ
e": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "s
tring"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}},
"type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}}, "type":
"array"}}, "fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH",
"RFM", "TMD"], "type": "string"}}, "fault_color": {"nullable": false, "required": true, "schema": {"type"
: "null"}}, "fault_name": {"nullable": false, "required": true, "schema": {"type": "null"}}, "impact_scop
e": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"check_re
quired": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fiel
ds": {"kind": {"nullable": false, "required": true, "schema": {"enum": ["CHAMBER", "LOT", "PARAMETER", "P
ROCESS_STEP", "SIBLING_CHAMBER", "WAFER"], "type": "string"}}, "relation": {"nullable": true, "required":
false, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min
_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "direct": {"nullable": false, "re
quired": true, "schema": {"items": {"additional_properties": false, "fields": {"kind": {"nullable": false
, "required": true, "schema": {"enum": ["CHAMBER", "LOT", "PARAMETER", "PROCESS_STEP", "SIBLING_CHAMBER",
"WAFER"], "type": "string"}}, "relation": {"nullable": true, "required": false, "schema": {"type": "stri
ng"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}}
, "type": "object"}, "type": "array"}}, "graph_conflict": {"nullable": false, "required": false, "schema":
{"default": false, "type": "boolean"}}, "reason_code": {"nullable": true, "required": false, "schema":
{"pattern": "[A-Z0-9_]+$", "type": "string"}, "status": {"nullable": false, "required": true, "schema":
{"enum": ["AVAILABLE", "EMPTY"], "type": "string"}}, "summary": {"nullable": true, "required": false, "s
chema": {"max_length": 2000, "type": "string"}}, "type": "object"}}, "latency_ms": {"nullable": false, "
required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "llm_model": {"nullable": false, "requir
ed": true, "schema": {"min_length": 1, "type": "string"}}, "post_action_observation": {"nullable": false,
"required": true, "schema": {"additional_properties": false, "fields": {"message": {"nullable": false, "
required": false, "schema": {"default": "\ucd5cluc885 \uc815luc801 \ub370luc774\ud130luc14bluc5d0lub294 \
uc870luc58 \uc774luc6d4 \uacf5luc815 \uad00luc21\uac12luc774 \luc5c6luc5b4 \ud6a8luacfclub97c \ud3c9luac
00lud560 \uc218 \luc5c6luc74c", "type": "string"}}, "status": {"nullable": false, "required": false, "sch
ema": {"default": "NOT_AVAILABLE_STATIC_DATASET", "type": "string"}}, "type": "object"}}, "predicted_faul
t_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "ty
pe": "string"}}, "prediction": {"nullable": true, "required": true, "schema": {"additional_properties": f
alse, "fields": {"cause_summary": {"nullable": false, "required": true, "schema": {"max_length": 2000, "m
in_length": 1, "type": "string"}}, "confidence": {"nullable": false, "required": true, "schema": {"maximu
m": 1.0, "minimum": 0.0, "type": "number"}}, "generated_at": {"nullable": false, "required": true, "schem
a": {"format": "date-time", "type": "string"}}, "input_tokens": {"nullable": false, "required": true, "sc
hema": {"minimum": 0.0, "type": "integer"}}, "llm_model": {"nullable": false, "required": true, "schema":
{"min_length": 1, "type": "string"}}, "output_tokens": {"nullable": false, "required": true, "schema": {
"minimum": 0.0, "type": "integer"}}, "predicted_fault_code": {"nullable": false, "required": true, "schem
a": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "prompt_version": {"nullable": fals
e, "required": true, "schema": {"min_length": 1, "type": "string"}}, "supporting_alerts": {"nullable": fa
```

```
lse, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}}, "type": "object"}, "type": "array"}}, "supporting_chunk_ids": {"nullable": false, "required": true, "schema": {"items": {"min_length": 1, "type": "string", "type": "array"}}, "supporting_relation_ids": {"nullable": false, "required": true, "schema": {"items": {"min_length": 1, "type": "string", "type": "array"}}, "uncertainty": {"nullable": false, "required": true, "schema": {"max_length": 1000, "type": "string"}}, "type": "object"}}, "recommended_action": {"nullable": true, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "similar_incidents": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "parameter_jaccard": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "predicted_fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "recommended_action": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "score": {"nullable": false, "required": true, "schema": {"maximum": 100.0, "minimum": 0.0, "type": "integer"}}}, "type": "object"}}, "type": "array"}}, "label": {"nullable": false, "required": false, "schema": {"default": "\u0000\u0015\u002d\u005f\u0037\u0074\u00d1\u00b4\u00e4\u0050\u00ac\u00ac\u00c0", "type": "string"}}, "reason_code": {"nullable": true, "required": false, "schema": {"pattern": "^[A-Z0-9_]+$", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["AVAILABLE", "EMPTY"], "type": "string"}}, "type": "object"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "tools": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"n": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "s": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}}, "type": "array"}}, "type": "object"}
AgentRunItem
{"additional_properties": false, "fields": {"action_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "approval_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "confidence": {"nullable": true, "required": true, "schema": {"maximum": 1, "minimum": 0, "type": "number"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object"}}, "type": "array"}}, "fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "fault_color": {"nullable": false, "required": true, "schema": {"type": "null"}}, "fault_name": {"nullable": false, "required": true, "schema": {"type": "null"}}, "latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "llm_model": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "predicted_fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "recommended_action": {"nullable": true, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "tools": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"n": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "s": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}}, "type": "array"}}, "type": "object"}
AlarmAskEvidence
{"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}
AlarmEvidence
{"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["ALARM"], "type": "string"}}, "type": "object"}
AlarmItem
```

```
{
  "additional_properties": false,
  "fields": {
    "action_code": {
      "nullable": true,
      "required": true,
      "schema": {
        "enum": ["EQP_HOLD", "MONITORING", "WARNING"],
        "type": "string"
      },
      "alarm_id": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
        "alarm_type": {
          "nullable": false,
          "required": true,
          "schema": {
            "enum": ["OOC", "OOS"],
            "type": "string"
          },
          "area": {
            "nullable": false,
            "required": true,
            "schema": {
              "enum": ["Etch", "Photo"],
              "type": "string"
            },
            "chamber": {
              "nullable": false,
              "required": true,
              "schema": {
                "min_length": 1,
                "type": "string"
              },
              "chamber_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
                "cl": {
                  "nullable": true,
                  "required": true,
                  "schema": {
                    "type": "number"
                  },
                  "equipment": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "min_length": 1,
                      "type": "string"
                    },
                    "equipment_id": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "min_length": 1,
                        "type": "string"
                      },
                      "fault": {
                        "nullable": true,
                        "required": true,
                        "schema": {
                          "enum": ["FOC", "MFD", "OTH", "RFM", "TMD"],
                          "type": "string"
                        },
                        "lcl": {
                          "nullable": true,
                          "required": true,
                          "schema": {
                            "type": "number"
                          },
                        "lot": {
                          "nullable": false,
                          "required": true,
                          "schema": {
                            "min_length": 1,
                            "type": "string"
                          },
                        "lot_id": {
                          "nullable": false,
                          "required": true,
                          "schema": {
                            "min_length": 1,
                            "type": "string"
                          },
                        "mes": {
                          "nullable": false,
                          "required": true,
                          "schema": {
                            "type": "string"
                          },
                        "mes_status": {
                          "nullable": true,
                          "required": true,
                          "schema": {
                            "enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"],
                            "type": "string"
                          },
                        "notify": {
                          "nullable": false,
                          "required": true,
                          "schema": {
                            "type": "boolean"
                          },
                        "notify_status": {
                          "nullable": true,
                          "required": true,
                          "schema": {
                            "enum": ["FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"],
                            "type": "string"
                          },
                        "occurred_at": {
                          "nullable": false,
                          "required": true,
                          "schema": {
                            "format": "date-time",
                            "type": "string"
                          },
                        "parameter": {
                          "nullable": false,
                          "required": true,
                          "schema": {
                            "min_length": 1,
                            "type": "string"
                          },
                        "parameter_id": {
                          "nullable": false,
                          "required": true,
                          "schema": {
                            "min_length": 1,
                            "type": "string"
                          },
                        "predicted_fault_code": {
                          "nullable": true,
                          "required": true,
                          "schema": {
                            "enum": ["FOC", "MFD", "OTH", "RFM", "TMD"],
                            "type": "string"
                          },
                        "recipe": {
                          "nullable": false,
                          "required": true,
                          "schema": {
                            "min_length": 1,
                            "type": "string"
                          },
                        "recipe_id": {
                          "nullable": false,
                          "required": true,
                          "schema": {
                            "min_length": 1,
                            "type": "string"
                          },
                        "recipe_step_no": {
                          "nullable": false,
                          "required": true,
                          "schema": {
                            "minimum": 1,
                            "type": "integer"
                          },
                        "rule_code": {
                          "nullable": false,
                          "required": true,
                          "schema": {
                            "enum": ["R03_CONSEC", "SUMMARY_OOC", "TRACE_OOS"],
                            "type": "string"
                          },
                        "seq_no": {
                          "nullable": true,
                          "required": true,
                          "schema": {
                            "minimum": 0,
                            "type": "integer"
                          },
                        "source": {
                          "nullable": false,
                          "required": true,
                          "schema": {
                            "enum": ["R03", "SUMMARY", "TRACE"],
                            "type": "string"
                          },
                        "statistic_type": {
                          "nullable": true,
                          "required": true,
                          "schema": {
                            "type": "string"
                          },
                        "step_no": {
                          "nullable": false,
                          "required": true,
                          "schema": {
                            "minimum": 1,
                            "type": "integer"
                          },
                        "ucl": {
                          "nullable": true,
                          "required": true,
                          "schema": {
                            "type": "number"
                          },
                        "value": {
                          "nullable": true,
                          "required": true,
                          "schema": {
                            "type": "number"
                          },
                        "wafer": {
                          "nullable": false,
                          "required": true,
                          "schema": {
                            "min_length": 1,
                            "type": "string"
                          },
                        "wafer_id": {
                          "nullable": false,
                          "required": true,
                          "schema": {
                            "min_length": 1,
                            "type": "string"
                          }
                        },
                        "rules": [
                          "source TRACE requires rule_code T",
                          "RACE_OOS",
                          "source SUMMARY requires rule_code SUMMARY_OOC",
                          "source R03 requires alarm_type OOS, rule_code R03_CONSEC, and value null",
                          "compatibility aliases derive only from canonical fields"
                        ],
                        "type": "object"
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

AlarmRef

```
{
  "additional_properties": false,
  "fields": {
    "alarm_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      },
    "source": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": ["R03", "SUMMARY", "TRACE"],
        "type": "string"
      }
    }
  },
  "type": "object"
}
```

AlarmSource

```
{
  "enum": ["R03", "SUMMARY", "TRACE"],
  "type": "string"
}
```

AnalysisQueryRequest

```
{
  "additional_properties": false,
  "fields": {
    "question": {
      "nullable": false,
      "required": true,
      "schema": {
        "max_length": 1000,
        "min_length": 1,
        "type": "string"
      }
    }
  },
  "type": "object"
}
```

AnalysisQueryResponse

```
{
  "additional_properties": false,
  "fields": {
    "columns": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "string"
      },
      "type": "array"
    },
    "cross_check": {
      "nullable": true,
      "required": false,
      "schema": {
        "additional_properties": false,
        "fields": {
          "cypher": {
            "nullable": true,
            "required": false,
            "schema": {
              "type": "string"
            },
            "status": {
              "nullable": false,
              "required": true,
              "schema": {
                "enum": ["MATCH", "MISMATCH", "SKIPPED"],
                "type": "string"
              },
            "summary": {
              "nullable": true,
              "required": false,
              "schema": {
                "type": "string"
              }
            },
            "type": "object"
          },
          "error_msg": {
            "nullable": true,
            "required": false,
            "schema": {
              "type": "string"
            },
            "generated_sql": {
              "nullable": true,
              "required": false,
              "schema": {
                "type": "string"
              },
            "group_by": {
              "nullable": false,
              "required": true,
              "schema": {
                "items": {
                  "type": "string"
                },
                "type": "array"
              },
            "is_rejected": {
              "nullable": false,
              "required": true,
              "schema": {
                "type": "boolean"
              },
            "is_valid": {
              "nullable": false,
              "required": true,
              "schema": {
                "type": "boolean"
              },
            "latency_ms": {
              "nullable": false,
              "required": true,
              "schema": {
                "minimum": 0.0,
                "type": "integer"
              },
            "metric": {
              "nullable": true,
              "required": false,
              "schema": {
                "additional_properties": false,
                "fields": {
                  "column": {
                    "nullable": true,
                    "required": false,
                    "schema": {
                      "type": "string"
                    },
                    "p": {
                      "nullable": true,
                      "required": false,
                      "schema": {
                        "maximum": 100.0,
                        "minimum": 0.0,
                        "type": "number"
                      },
                    "type": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "enum": ["count", "max", "mean", "median", "min", "percentile", "ratio", "std", "sum"],
                        "type": "string"
                      }
                    }
                  },
                  "metric_result": {
                    "nullable": false,
                    "required": false,
                    "schema": {
                      "type": "union",
                      "variants": [
                        {
                          "type": "integer"
                        },
                        {
                          "type": "number"
                        }
                      ],
                      "items": {
                        "additional_properties": false,
                        "fields": {
                          "group": {
                            "nullable": false,
                            "required": true,
                            "schema": {
                              "additional_properties": true,
                              "fields": {},
                              "type": "object"
                            },
                            "value": {
                              "nullable": false,
                              "required": false,
                              "schema": {
                                "type": "union",
                                "variants": [
                                  {
                                    "type": "integer"
                                  },
                                  {
                                    "type": "number"
                                  },
                                  {
                                    "type": "null"
                                  }
                                ]
                              }
                            },
                            "nl_query_log_id": {
                              "nullable": true,
                              "required": false,
                              "schema": {
                                "minimum": 1.0,
                                "type": "integer"
                              }
                            },
                            "question": {
                              "nullable": false,
                              "required": true,
                              "schema": {
                                "type": "string"
                              },
                            "reject_reason": {
                              "nullable": true,
                              "required": false,
                              "schema": {
                                "type": "string"
                              },
                            "row_count": {
                              "nullable": false,
                              "required": true,
                              "schema": {
                                "minimum": 0.0,
                                "type": "integer"
                              },
                            "rows": {
                              "nullable": false,
                              "required": true,
                              "schema": {
                                "items": {
                                  "additional_properties": true,
                                  "fields": {},
                                  "type": "object"
                                },
                                "type": "array"
                              },
                            "visualization": {
                              "nullable": true,
                              "required": false,
                              "schema": {
                                "additional_properties": false,
                                "fields": {
                                  "chart_type": {
                                    "nullable": false,
                                    "required": true,
                                    "schema": {
                                      "enum": ["bar", "histogram", "line", "table"],
                                      "type": "string"
                                    },
                                  "x": {
                                    "nullable": true,
                                    "required": false,
                                    "schema": {
                                      "type": "string"
                                    },
                                  "y": {
                                    "nullable": true,
                                    "required": false,
                                    "schema": {
                                      "type": "string"
                                    }
                                  }
                                }
                              }
                            }
                          }
                        }
                      ]
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```



```
{ "nullable": true, "required": false, "schema": { "type": "string"}}, "type": "object"}], "type": "object"}  
ApprovalDecisionRequest  
{ "additional_properties": false, "fields": {"decided_by": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "decision": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "REJECTED"], "type": "string"}}, "decision_comment": {"nullable": true, "required": false, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}}, "type": "object"}  
ApprovalsItem  
{ "additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approved_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "approved_by": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "decision_comment": {"nullable": true, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "equipment": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "predicted_fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "rules": [{"PE_NDING requires all decision fields null", "APPROVED or REJECTED requires decided_by and decided_at", "only EQP_HOLD actions have approvals"}, {"type": "object"}]}  
AskDocumentEvidenceAlias  
{ "additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "doc_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}}, "type": "object"}  
AskToolItem  
{ "additional_properties": false, "fields": {"name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}  
AuditLogItem  
{ "additional_properties": false, "fields": {"actor": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "actor_id": {"nullable": true, "required": true, "schema": {"type": "string"}}, "actor_type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "after": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "audit_id": {"nullable": false, "required": true, "schema": {"minimum": 1, "type": "integer"}}, "before": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "detail": {"nullable": true, "required": true, "schema": {"type": "string"}}, "entity": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "entity_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "event": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "event_type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "type": "object"}  
AuditLogPageResponse  
{ "additional_properties": true, "fields": {"event_type_counts": {"nullable": false, "required": true, "schema": {"additional_properties": {"type": "integer"}, "fields": {}, "type": "object"}, "event_types": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": true, "fields": {"actor": {"nullable": false, "required": true, "schema": {"type": "string"}}, "actor_id": {"nullable": true, "required": true, "schema": {"type": "string"}}, "actor_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "after": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "audit_id": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "before": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "detail": {"nullable": true, "required": true, "schema": {"type": "string"}}, "entity": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "fields": {"type": "integer"}, "fi
```

```

string"}, {"nullable": false, "required": true, "schema": {"type": "string"}}, {"occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}}, {"type": "object"}, {"type": "array"}], "page": {"nullable": false, "required": true, "schema": {"type": "integer"}}, {"size": {"nullable": false, "required": true, "schema": {"type": "integer"}}, {"total": {"nullable": false, "required": true, "schema": {"type": "integer"}}}, {"type": "object"}
AutoToolCallItem
{"additional_properties": false, "fields": {"n": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"s": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, {"status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, {"tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}}, {"type": "object"}
ChamberGraphResponse
{"additional_properties": false, "fields": {"context": {"nullable": false, "required": true, "schema": {"type": "string"}}, {"adjacent_process_step_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, {"area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, {"chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"model_code": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"parameter_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, {"process_step_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"relation_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, {"graph_revision": {"nullable": false, "required": true, "schema": {"pattern": "^\\[0-9-a-f\\{64\\}$", "type": "string"}}, {"node_count": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, {"nodes": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"business_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"label": {"nullable": false, "required": true, "schema": {"enum": ["Area", "Chamber", "Equipment"], "EquipmentModel": "Parameter"}, "ProcessStep"], "type": "string"}}, {"name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"properties": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, {"type": "object"}}, {"relationship_count": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, {"relationships": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"from_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"relation_id": {"nullable": false, "required": true, "schema": {"pattern": "^REL-[0-9-a-f\\{20\\}$", "type": "string"}}, {"to_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"type": "object"}}, {"type": "array"}}, {"rules": {"node_count_equals len(nodes)", "relationship_count_equals len(relationships)"}, {"type": "object"}
ChamberRelationResponse
{"additional_properties": false, "fields": {"graph_revision": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"nodes": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"business_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"display_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"label": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"properties": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, {"type": "object"}}, {"type": "array"}}, {"relationships": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"source": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"target": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"root_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"type": "object"}}, {"type": "array"}}, {"type": "object"}
ChartType
{"enum": ["bar", "histogram", "line", "table"], "type": "string"}
ChatToolCallItem
{"additional_properties": false, "fields": {"name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"result": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, {"tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"type": "object"}
CrossCheck
{"additional_properties": false, "fields": {"cypher": {"nullable": true, "required": false, "schema": {"type": "string"}}, {"status": {"nullable": false, "required": true, "schema": {"enum": ["MATCH", "MISMATCH", "SKIPPED"], "type": "string"}}, {"summary": {"nullable": true, "required": false, "schema": {"type": "string"}}, {"type": "object"}
DeliveryCallbackRequest
{"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES MOCK"], "type": "string"}}, {"completed_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, {"error_code": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, {"type": "string"}}, {"type": "object"}

```

```
schema": {"min_length": 1, "type": "string"}, "event_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "provider_message_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}, "request_hash": {"nullable": false, "required": true, "schema": {"max_length": 64, "min_length": 64, "pattern": "^[0-9a-f]{64}$", "type": "string"}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAILED", "SENT"], "type": "string"}}}, "rules": ["SENT requires provider_message_id non-null and error_code null", "FAILED requires error_code non-null and permits provider_message_id null"], "type": "object"}
DeliveryChannel
{"enum": ["EMAIL", "MES MOCK"], "type": "string"}
DeliveryResult
{"additional_properties": false, "fields": {"action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES MOCK"], "type": "string"}, "completed_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}, "duplicate": {"nullable": false, "required": true, "schema": {"type": "boolean"}, "error_code": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}, "provider_message_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}, "request_hash": {"nullable": false, "required": true, "schema": {"max_length": 64, "min_length": 64, "pattern": "^[0-9a-f]{64}$", "type": "string"}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAILED", "SENT"], "type": "string"}, "rules": ["SENT requires provider_message_id non-null and error_code null", "FAILED requires error_code non-null and permits provider_message_id null"], "type": "object"}
DeliveryStatus
{"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}
DiagnosisBlock
{"additional_properties": false, "fields": {"alternative_hypotheses": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"lower_rank_reason": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}, "summary": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}, "cause_summary": {"nullable": true, "required": false, "schema": {"max_length": 2000, "min_length": 1, "type": "string"}, "confidence": {"nullable": true, "required": false, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}, "diagnostic_coverage": {"nullable": true, "required": false, "schema": {"type": "string"}, "evidence_synthesis": {"nullable": true, "required": false, "schema": {"max_length": 2000, "type": "string"}, "limitations": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}, "observations": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}, "predicted_fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}, "reason_code": {"nullable": true, "required": false, "schema": {"pattern": "^[A-Z0-9_]+$", "type": "string"}, "status": {"nullable": false, "required": true, "schema": {"enum": ["AVAILABLE", "EMPTY"], "type": "string"}, "verification_steps": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "type": "object"}
DocumentAskEvidence
{"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}
DocumentChunkItem
{"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "chunk_seq": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}, "content": {"nullable": false, "required": true, "schema": {"type": "string"}, "section_title": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}
DocumentDetailResponse
{"additional_properties": false, "fields": {"chunks": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "chunk_seq": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}, "content": {"nullable": false, "required": true, "schema": {"type": "string"}, "section_title": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}, "doc_type": {"nullable": true, "required": false, "schema": {"enum": ["MANUAL", "SPEC", "TROUBLESHOOT"], "type": "string"}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}, "source_path": {"nullable": true, "required": false, "schema": {"type": "string"}, "title": {"nullable": false, "required": true, "schema": {"type": "string"}, "version": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}
DocumentEvidence
{"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}, "type": {"nullable": false, "required": true, "schema": {"enum": ["DOCUMENT"], "type": "string"}}, "type": "object"}}
```

DocumentHit

```
{
  "additional_properties": false,
  "fields": {
    "chunk_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "pattern": "^.+cs2:[0-9]{4}$",
        "type": "string"
      }
    },
    "content": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "doc_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "document_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "model_code": {
      "nullable": true,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "score": {
      "nullable": false,
      "required": true,
      "schema": {
        "maximum": 1,
        "minimum": -1,
        "type": "number"
      }
    },
    "section": {
      "nullable": true,
      "required": true,
      "schema": {
        "type": "string"
      }
    },
    "title": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    }
  },
  "type": "object"
}
```

DocumentSearchRequest

```
{
  "additional_properties": false,
  "fields": {
    "model_code": {
      "nullable": true,
      "required": false,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "query": {
      "nullable": false,
      "required": true,
      "schema": {
        "max_length": 1000,
        "min_length": 1,
        "type": "string"
      }
    },
    "top_k": {
      "nullable": false,
      "required": false,
      "schema": {
        "default": 4,
        "maximum": 10,
        "minimum": 1,
        "type": "integer"
      }
    }
  },
  "type": "object"
}
```

DocumentType

```
{
  "enum": ["MANUAL", "SPEC", "TROUBLESHOOT"],
  "type": "string"
}
```

ErrorCode

```
{
  "enum": [
    "APPROVAL_ALREADY_DECIDED",
    "DEPENDENCY_NOT_READY",
    "IDEMPOTENCY_CONFLICT",
    "INCIDENT_ALREADY_PROCESSED",
    "INCIDENT_ALREADY_RUNNING",
    "INTERNAL_ERROR",
    "LEGACY_APPROVAL_NOT_LINKED",
    "LLM_NOT_READY",
    "MODEL_NOT_READY",
    "POLICY_REJECTED",
    "RESOURCE_NOT_FOUND",
    "UNAUTHORIZED",
    "VALIDATION_ERROR"
  ],
  "type": "string"
}
```

ErrorResponse

```
{
  "additional_properties": false,
  "fields": {
    "code": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "details": {
      "nullable": false,
      "required": true,
      "schema": {
        "additional_properties": true,
        "fields": {},
        "type": "object"
      }
    },
    "message": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    }
  },
  "type": "object"
}
```

EvaluationListResponse

```
{
  "additional_properties": false,
  "fields": {
    "items": {
      "nullable": false,
      "required": true,
      "schema": {
        "items": {
          "additional_properties": false,
          "fields": {
            "accuracy": {
              "nullable": false,
              "required": true,
              "schema": {
                "maximum": 1.0,
                "minimum": 0.0,
                "type": "number"
              }
            },
            "correct": {
              "nullable": false,
              "required": true,
              "schema": {
                "minimum": 0,
                "type": "integer"
              }
            },
            "defense_passed": {
              "nullable": false,
              "required": true,
              "schema": {
                "minimum": 0,
                "type": "integer"
              }
            },
            "defense_total": {
              "nullable": false,
              "required": true,
              "schema": {
                "minimum": 0,
                "type": "integer"
              }
            },
            "executed_at": {
              "nullable": false,
              "required": true,
              "schema": {
                "format": "date-time",
                "type": "string"
              }
            },
            "items": {
              "nullable": false,
              "required": true,
              "schema": {
                "items": {
                  "additional_properties": false,
                  "fields": {
                    "actual_result": {
                      "nullable": true,
                      "required": false,
                      "schema": {
                        "type": "any"
                      }
                    },
                    "actual_visualization": {
                      "nullable": true,
                      "required": false,
                      "schema": {
                        "additional_properties": false,
                        "fields": {
                          "chart_type": {
                            "nullable": false,
                            "required": true,
                            "schema": {
                              "enum": ["bar", "histogram", "line", "table"],
                              "type": "string"
                            }
                          },
                          "x": {
                            "nullable": true,
                            "required": false,
                            "schema": {
                              "type": "string"
                            }
                          },
                          "y": {
                            "nullable": true,
                            "required": false,
                            "schema": {
                              "type": "string"
                            }
                          }
                        },
                        "type": "object"
                      }
                    },
                    "attempt_count": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "minimum": 0,
                        "type": "integer"
                      }
                    },
                    "case_id": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "min_length": 1,
                        "type": "string"
                      }
                    },
                    "case_type": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "enum": ["DEFENSE", "GOLD"],
                        "type": "string"
                      }
                    },
                    "expected_result": {
                      "nullable": true,
                      "required": false,
                      "schema": {
                        "type": "any"
                      }
                    },
                    "expected_visualization": {
                      "nullable": true,
                      "required": false,
                      "schema": {
                        "additional_properties": false,
                        "fields": {
                          "chart_type": {
                            "nullable": false,
                            "required": true,
                            "schema": {
                              "enum": ["bar", "histogram", "line", "table"],
                              "type": "string"
                            }
                          },
                          "x": {
                            "nullable": true,
                            "required": false,
                            "schema": {
                              "type": "string"
                            }
                          },
                          "y": {
                            "nullable": true,
                            "required": false,
                            "schema": {
                              "type": "string"
                            }
                          }
                        },
                        "type": "object"
                      }
                    },
                    "generated_sql": {
                      "nullable": true,
                      "required": false,
                      "schema": {
                        "type": "string"
                      }
                    },
                    "latency_ms": {
                      "nullable": true,
                      "required": false,
                      "schema": {
                        "minimum": 0,
                        "type": "integer"
                      }
                    },
                    "passed": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "type": "boolean"
                      }
                    },
                    "question": {
                      "nullable": true,
                      "required": false,
                      "schema": {
                        "type": "string"
                      }
                    },
                    "reason": {
                      "nullable": true,
                      "required": false,
                      "schema": {
                        "type": "string"
                      }
                    },
                    "type": "object",
                    "type": "array"
                  },
                  "model": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "min_length": 1,
                      "type": "string"
                    }
                  },
                  "prompt_version": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "min_length": 1,
                      "type": "string"
                    }
                  },
                  "provider": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "min_length": 1,
                      "type": "string"
                    }
                  },
                  "run_id": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "min_length": 1,
                      "type": "string"
                    }
                  },
                  "temperature": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "type": "number"
                    }
                  },
                  "total": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "minimum": 0,
                      "type": "integer"
                    }
                  },
                  "type": "object",
                  "type": "array"
                },
                "page": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "minimum": 1,
                    "type": "integer"
                  }
                },
                "size": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "maximum": 100,
                    "minimum": 1,
                    "type": "integer"
                  }
                },
                "total": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "minimum": 0,
                    "type": "integer"
                  }
                }
              }
            }
          },
          "type": "object"
        }
      },
      "type": "array"
    }
  },
  "type": "object"
}
```

EvidenceAssessmentBlock

```
{
  "additional_properties": false,
  "fields": {
    "available_sources": {
      "nullable": false,
      "required": true,
      "schema": {
        "items": {
          "enum": ["FDC", "GRAPH", "POSTGRES_ROUTE", "RAG"],
          "type": "string"
        },
        "type": "array"
      }
    },
    "conflicting_source_ids": {
      "nullable": false,
      "required": true,
      "schema": {
        "items": {
          "type": "string"
        },
        "type": "array"
      }
    },
    "missing_sources": {
      "nullable": false,
      "required": true,
      "schema": {
        "items": {
          "enum": ["FDC", "GRAPH", "POSTGRES_ROUTE", "RAG"],
          "type": "string"
        },
        "type": "array"
      }
    },
    "reason_codes": {
      "nullable": false,
      "required": true,
      "schema": {
        "items": {
          "type": "string"
        },
        "type": "array"
      }
    },
    "status": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": ["CONFLICT", "EMPTY", "PARTIAL", "SUFFICIENT"],
        "type": "string"
      }
    }
  },
  "type": "object"
}
```

EvidenceItem

```
{
  "discriminator": {
    "type": "type",
    "type": "discriminated_union",
    "variants": {
      "ALARM": {
        "additional_properties": false,
        "fields": {
          "excerpt": {
            "nullable": false,
            "required": true,
            "schema": {
              "type": "string"
            }
          },
          "source": {
            "nullable": false,
            "required": true,
            "schema": {
              "type": "string"
            }
          }
        }
      }
    }
  }
}
```

```
_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["ALARM"], "type": "string"}}, "type": {"object"}, "DOCUMENT": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["DOCUMENT"], "type": "string"}}, "type": {"object"}, "GRAPH": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "graph_revision": {"nullable": false, "required": true, "schema": {"pattern": "^0-9a-f{64}$", "type": "string"}}, "relation_id": {"nullable": false, "required": true, "schema": {"pattern": "^REL-[0-9a-f]{20}$", "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["GRAPH"], "type": "string"}}, "type": {"object"}, "METROLOGY": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["METROLOGY"], "type": "string"}}, "rules": ["metrology.alarms_result is forbidden"], "type": {"object"}, "TRACE": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["TRACE"], "type": "string"}}, "type": {"object"}}}
```

FaultHypothesis

```
{"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}
```

GraphAskEvidence

```
{"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "graph_revision": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "relation_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"object"}}
```

GraphContext

```
{"additional_properties": false, "fields": {"adjacent_process_step_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "model_code": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "parameter_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "process_step_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "relation_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "type": {"object"}}
```

GraphEvidence

```
{"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "graph_revision": {"nullable": false, "required": true, "schema": {"pattern": "^0-9a-f{64}$", "type": "string"}}, "relation_id": {"nullable": false, "required": true, "schema": {"pattern": "^REL-[0-9a-f]{20}$", "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["GRAPH"], "type": "string"}}, "type": {"object"}}
```

GraphNode

```
{"additional_properties": false, "fields": {"business_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "label": {"nullable": false, "required": true, "schema": {"enum": ["Area", "Chamber", "Equipment", "EquipmentModel", "Parameter", "ProcessStep"], "type": "string"}}, "name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "properties": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": {"object"}}, "type": {"object"}}
```

GraphQueryRequest

```
{"additional_properties": false, "fields": {"question": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": {"object"}}
```

GraphQueryResponse

```
{"additional_properties": false, "fields": {"columns": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "error_msg": {"nullable": true, "required": false, "schema": {"type": "string"}}, "generated_cypher": {"nullable": true, "required": false, "schema": {"type": "string"}}, "is_rejected": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "is_valid": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "question": {"nullable": false, "required": true, "schema": {"type": "string"}}, "reject_reason": {"nullable": true, "required": false, "schema": {"type": "string"}}, "row_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}
```

```
"integer"}}, "rows": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": true, "fields": {}, "type": "object"}, "type": "array"}}, "type": "object"}
GraphRelationship
{"additional_properties": false, "fields": {"from_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "relation_id": {"nullable": false, "required": true, "schema": {"pattern": "^REL-[0-9a-f]{20}$", "type": "string"}}, "to_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}}, "type": "object"}
GroupedMetricResult
{"additional_properties": false, "fields": {"group": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "value": {"nullable": false, "required": false, "schema": {"type": "union", "variants": [{"type": "integer"}, {"type": "number"}, {"type": "null"}]}}, "type": "object"}
HTTPValidationError
{"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}, "message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}}, "type": "object"}, "type": "array"}}, "type": "object"}
HealthResponse
{"additional_properties": false, "fields": {"status": {"nullable": false, "required": true, "schema": {"enum": ["UP"], "type": "string"}}, "type": "object"}
ImpactScopeBlock
{"additional_properties": false, "fields": {"check_required": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"kind": {"nullable": false, "required": true, "schema": {"enum": ["CHAMBER", "LOT", "PARAMETER", "PROCESS_STEP", "PROCESS_STEP", "SIBLING_CHAMBER", "WAFER"], "type": "string"}}, "relation": {"nullable": true, "required": false, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}, "direct": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"kind": {"nullable": false, "required": true, "schema": {"enum": ["CHAMBER", "LOT", "PARAMETER", "PROCESS_STEP", "SIBLING_CHAMBER", "WAFER"], "type": "string"}}, "relation": {"nullable": true, "required": false, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}, "graph_conflict": {"nullable": false, "required": false, "schema": {"default": false, "type": "boolean"}}, "reason_code": {"nullable": true, "required": false, "schema": {"pattern": "^[A-Z0-9_]+$", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["AVAILABLE", "EMPTY"], "type": "string"}}, "summary": {"nullable": true, "required": false, "schema": {"max_length": 2000, "type": "string"}}, "type": "object"}
ImpactScopeItem
{"additional_properties": false, "fields": {"kind": {"nullable": false, "required": true, "schema": {"enum": ["CHAMBER", "LOT", "PARAMETER", "PROCESS_STEP", "SIBLING_CHAMBER", "WAFER"], "type": "string"}}, "relation": {"nullable": true, "required": false, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
MetricPlan
{"additional_properties": false, "fields": {"column": {"nullable": true, "required": false, "schema": {"type": "string"}}, "p": {"nullable": true, "required": false, "schema": {"maximum": 100.0, "minimum": 0.0, "type": "number"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["count", "max", "mean", "median", "min", "percentile", "ratio", "std", "sum"], "type": "string"}}, "type": "object"}
MetrologyAskEvidence
{"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}
MetrologyEvidence
{"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["METROLOGY"], "type": "string"}}, "rules": ["metrology.alarm_result is forbidden"], "type": "object"}
NIQueryHistoryResponse
{"additional_properties": false, "fields": {"items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"asked_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "error_msg": {"nullable": true, "required": false, "schema": {"type": "string"}}, "generated_sql": {"nullable": true, "required": false, "schema": {"type": "string"}}, "is_rejected": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "is_valid": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "latency_ms": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "nl_query_log_id": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "outcome": {"nullable": false, "required": true, "schema": {"enum": ["DB_ERROR", "POLICY_REJECTED", "SUCCESS", "VALIDATION_FAILED"], "type": "string"}}, "question": {"nullable": false, "required": true, "schema": {"type": "string"}}, "reject_reason": {"nullable": true, "required": false, "schema": {"type": "string"}}, "row_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}, "type": "array"}, "page": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "size": {"nullab
```

```
le": false, "required": true, "schema": {"maximum": 100.0, "minimum": 1.0, "type": "integer"}}, "total":
{"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}
NIQueryLogItem
{"additional_properties": false, "fields": {"asked_at": {"nullable": false, "required": true, "schema": {
"format": "date-time", "type": "string"}}, "error_msg": {"nullable": true, "required": false, "schema": {
"type": "string"}}, "generated_sql": {"nullable": true, "required": false, "schema": {"type": "string"}},
"is_rejected": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "is_valid": {"nulla
ble": false, "required": true, "schema": {"type": "boolean"}}, "latency_ms": {"nullable": true, "required
": false, "schema": {"minimum": 0.0, "type": "integer"}}, "nl_query_log_id": {"nullable": false, "require
d": true, "schema": {"minimum": 1.0, "type": "integer"}}, "outcome": {"nullable": false, "required": true
, "schema": {"enum": ["DB_ERROR", "POLICY_REJECTED", "SUCCESS", "VALIDATION_FAILED"], "type": "string"}},
"question": {"nullable": false, "required": true, "schema": {"type": "string"}}, "reject_reason": {"null
able": true, "required": false, "schema": {"type": "string"}}, "row_cnt": {"nullable": true, "required":
false, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}
NIQueryOutcome
{"enum": ["DB_ERROR", "POLICY_REJECTED", "SUCCESS", "VALIDATION_FAILED"], "type": "string"}
ParameterItem
{"additional_properties": false, "fields": {"LCL": {"nullable": true, "required": true, "schema": {"type"
: "number"}}, "LSL": {"nullable": true, "required": true, "schema": {"type": "number"}}, "TARGET": {"null
able": false, "required": true, "schema": {"type": "number"}}, "UCL": {"nullable": false, "required": tru
e, "schema": {"type": "number"}}, "USL": {"nullable": false, "required": true, "schema": {"type": "number
"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"
}}, "ctrl_lower": {"nullable": true, "required": true, "schema": {"type": "number"}}, "ctrl_upper": {"nul
lable": false, "required": true, "schema": {"type": "number"}}, "name": {"nullable": false, "required": t
rue, "schema": {"min_length": 1, "type": "string"}}, "parameter_id": {"nullable": false, "required": true
, "schema": {"min_length": 1, "type": "string"}}, "parameter_name": {"nullable": false, "required": true,
"schema": {"min_length": 1, "type": "string"}}, "spec_lower": {"nullable": true, "required": true, "sche
ma": {"type": "number"}}, "spec_upper": {"nullable": false, "required": true, "schema": {"type": "number"
}}, "target_value": {"nullable": false, "required": true, "schema": {"type": "number"}}, "unit": {"nullab
le": true, "required": true, "schema": {"type": "string"}}, "upper_only": {"nullable": false, "required":
true, "schema": {"type": "boolean"}}, "type": "object"}
PostActionObservationBlock
{"additional_properties": false, "fields": {"message": {"nullable": false, "required": false, "schema": {
"default": "\ucd5c\uc885 \uc815\uc801 \ub370\uc774\ud130\uc14b\uc5d0\ub294 \uc870\uce58 \uc774\ud6c4 \uac
f5\uc815 \uad00\uce21\uac12\uc774 \uc5c6\uc5b4 \ud6a8\uacfc\ub97c \ud3c9\uac00\ud560 \uc218 \uc5c6\uc74c"
, "type": "string"}}, "status": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILAB
LE_STATIC_DATASET", "type": "string"}}, "type": "object"}
PublicAgentRunItem
{"additional_properties": false, "fields": {"action_id": {"nullable": true, "required": true, "schema": {
"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"mi
n_length": 1, "type": "string"}}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_lengt
h": 1, "type": "string"}}, "alarm_source": {"nullable": false, "required": true, "schema": {"enum": ["R03
", "SUMMARY", "TRACE"], "type": "string"}}, "approval_id": {"nullable": true, "required": true, "schema":
{"min_length": 1, "type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_l
ength": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length
": 1, "type": "string"}}, "confidence": {"nullable": true, "required": true, "schema": {"maximum": 1.0, "
minimum": 0.0, "type": "number"}}, "created_at": {"nullable": false, "required": true, "schema": {"format
": "date-time", "type": "string"}}, "deliveries": {"nullable": false, "required": true, "schema": {"items
": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema":
{"enum": ["EMAIL", "MES"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema":
{"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}},
"type": "object"}, "type": "array"}}, "fault_code": {"nullable": true, "required": true, "schema": {"en
um": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "fault_color": {"nullable": false, "require
d": true, "schema": {"type": "null"}}, "fault_name": {"nullable": false, "required": true, "schema": {"ty
pe": "null"}}, "latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "in
teger"}}, "llm_model": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"
}}, "predicted_fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH",
"RFM", "TMD"], "type": "string"}}, "recommended_action": {"nullable": true, "required": true, "schema":
{"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "status": {"nullable": false, "requir
ed": true, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}},
"tools": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fie
lds": {"n": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result
_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "s": {"n
ullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}},
"status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "ty
pe": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "
string"}}, "type": "object"}, "type": "array"}}, "type": "object"}
PublicApprovalDecision
{"enum": ["APPROVED", "REJECTED"], "type": "string"}
PublicApprovalItem
{"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema":
{"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "r
equired": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "requ
ired": true, "schema": {"min_length": 1, "type": "string"}}, "approval_id": {"nullable": false, "required
```

```
": true, "schema": {"min_length": 1, "type": "string"}}, "approved_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "approved_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "decision_comment": {"nullable": true, "required": true, "schema": {"type": "string"}}, "equipment": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "predicted_fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "type": "object"}
PublicApprovalStatus
{"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}
PublicDeliveryChannel
{"enum": ["EMAIL", "MES"], "type": "string"}
PublicDeliveryItem
{"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}}, "type": "object"}
PublicToolCallItem
{"additional_properties": false, "fields": {"n": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "s": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}}, "type": "object"}
ReadinessCheck
{"additional_properties": false, "fields": {"latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "reason_code": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAIL", "PASS"], "type": "string"}}}, "type": "object"}
ReadinessChecks
{"additional_properties": false, "fields": {"kafka": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "reason_code": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAIL", "PASS"], "type": "string"}}, "type": "object"}}, "n8n": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "reason_code": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAIL", "PASS"], "type": "string"}}, "type": "object"}}, "neo4j": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "reason_code": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAIL", "PASS"], "type": "string"}}, "type": "object"}}, "postgresql_runtime": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "reason_code": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAIL", "PASS"], "type": "string"}}, "type": "object"}}, "reference_migration": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "reason_code": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAIL", "PASS"], "type": "string"}}, "type": "object"}}, "type": "object"}
ReadinessResponse
{"additional_properties": false, "fields": {"checks": {"nullable": false, "required": true, "schema": {"a
```


[illegible]

SqlValidateResponse

```
{
  "additional_properties": false,
  "fields": {
    "checks": {
      "nullable": true,
      "required": false,
      "schema": {
        "items": {
          "additional_properties": false,
          "fields": {
            "key": {
              "nullable": false,
              "required": true,
              "schema": {
                "min_length": 1,
                "type": "string"
              }
            },
            "label": {
              "nullable": false,
              "required": true,
              "schema": {
                "min_length": 1,
                "type": "string"
              }
            },
            "ok": {
              "nullable": false,
              "required": true,
              "schema": {
                "type": "boolean"
              }
            },
            "type": {
              "object": true,
              "type": "array"
            },
            "normalized_sql": {
              "nullable": true,
              "required": false,
              "schema": {
                "type": "string"
              }
            },
            "reason": {
              "nullable": false,
              "required": true,
              "schema": {
                "type": "string"
              }
            },
            "valid": {
              "nullable": false,
              "required": true,
              "schema": {
                "type": "boolean"
              }
            },
            "type": "object"
          }
        }
      }
    }
  }
}
```

ToolCallStatus

```
{
  "enum": ["ERROR", "SUCCESS", "TIMEOUT"],
  "type": "string"
}
```

TraceAskEvidence

```
{
  "additional_properties": false,
  "fields": {
    "excerpt": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "source_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "title": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "type": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "string"
      }
    },
    "type": "object"
  }
}
```

TraceEvidence

```
{
  "additional_properties": false,
  "fields": {
    "excerpt": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "string"
      }
    },
    "source_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "title": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "type": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": ["TRACE"],
        "type": "string"
      }
    },
    "type": "object"
  }
}
```

TracePoint

```
{
  "additional_properties": false,
  "fields": {
    "measured_at": {
      "nullable": false,
      "required": true,
      "schema": {
        "format": "date-time",
        "type": "string"
      }
    },
    "recipe_step_no": {
      "nullable": false,
      "required": true,
      "schema": {
        "minimum": 1,
        "type": "integer"
      }
    },
    "seq_no": {
      "nullable": false,
      "required": true,
      "schema": {
        "minimum": 0,
        "type": "integer"
      }
    },
    "value": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "number"
      }
    },
    "type": "object"
  }
}
```

ValidationCheck

```
{
  "additional_properties": false,
  "fields": {
    "key": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "label": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "ok": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "boolean"
      }
    },
    "type": "object"
  }
}
```

ValidationError

```
{
  "additional_properties": true,
  "fields": {
    "loc": {
      "nullable": false,
      "required": true,
      "schema": {
        "items": {
          "type": "union",
          "variants": [
            {
              "type": "string"
            },
            {
              "type": "integer"
            }
          ]
        },
        "type": "array"
      }
    },
    "msg": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "string"
      }
    },
    "type": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "string"
      }
    },
    "type": "object"
  }
}
```

VisualizationPlan

```
{
  "additional_properties": false,
  "fields": {
    "chart_type": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": ["bar", "histogram", "line", "table"],
        "type": "string"
      }
    },
    "x": {
      "nullable": true,
      "required": false,
      "schema": {
        "type": "string"
      }
    },
    "y": {
      "nullable": true,
      "required": false,
      "schema": {
        "type": "string"
      }
    },
    "type": "object"
  }
}
```

AUDIT EVENTS

DETECTION_COMPLETED / LOT_HIST / A

AGENT_RUN_STARTED / AGENT_RUN / C

HYPOTHESIS_GENERATED / AGENT_RUN / C

APPROVAL_REQUESTED / APPROVAL / C

APPROVAL_DECIDED / APPROVAL / C

ACTION_SENT / ACTION / C delivery service

ACTION_SEND_FAILED / ACTION / C delivery service

AGENT_RUN_COMPLETED / AGENT_RUN / C

AGENT_RUN_FAILED / AGENT_RUN / C